# Advanced Research Project

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# News-Driven Electricity Trading: Augmenting Ensemble Models with NLP for Signal Generation in German Power Markets

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# Management Summary

This research project examines whether energy-related news can be transformed into statistically and economically meaningful trading signals for the German electricity market using modern Natural Language Processing (NLP) techniques. The study is motivated by the growing role of information shocks in short-term power price formation and the limited empirical evidence on the tradability of news-based forecasts in electricity markets.

The methodological framework combines zero-shot topic classification and transformer-based sentence embeddings to extract quantitative signals from energy-related news headlines. These news-derived features are integrated with price-based variables in a two-stage modelling architecture. An upstream XGBoost classifier generates probabilistic forecasts of directional price movements, which are subsequently converted into trading signals by a LightGBM meta-model. Model performance is evaluated both statistically, using out-of-sample classification metrics and formal significance testing, and economically, through a stylised portfolio backtest incorporating conservative transaction cost assumptions.

The results show that incorporating news-derived information leads to clear improvements over price-based and naïve long-only benchmarks. Statistically, the news-augmented signal model achieves higher accuracy and a stronger macro-averaged F1 score, with the performance difference confirmed as statistically significant. These gains are driven by improved balance across Long, Neutral, and Short trading outcomes rather than dominance in a single class. Economically, the news-informed strategy outperforms both benchmarks in cumulative returns and risk-adjusted performance while maintaining shallow, well-contained drawdowns. The naïve strategy, in contrast, experiences severe capital losses, underscoring the importance of directional timing in the German power market.

Overall, the findings demonstrate that energy-related news contains incremental information that can be systematically exploited within a structured trading framework. At the same time, the results should be interpreted as a proof of concept. The reliance on zero-shot classification, dimensionality reduction, and stylised execution assumptions introduces sensitivity to modelling choices and limits strict reproducibility. Further methodological refinement and robustness testing are therefore required before real-world deployment can be considered.

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# 1 Introduction

The present chapter provides the conceptual and structural foundation for the study. The initial section (1.1) provides a contextual foundation, while the subsequent section (1.2) outlines the central research question. The objectives of the research project and its academic significance are addressed in Section 1.3, followed by an outline of the research project's structure in Section 1.4.

## 1.1 Background

Energy constitutes a fundamental input to modern economic activity and social welfare. From an economic perspective, energy enables production, transportation, and consumption across virtually all sectors, while from a societal standpoint, it underpins living standards and technological development. Historically, macroeconomic growth has been closely associated with increasing energy consumption, reflecting the central role of energy in industrialisation and economic expansion (Stern, 2011, p. 26). In this sense, energy can be viewed not merely as a commodity, but as a prerequisite for organised economic life.

Within the broader concept of energy, electricity occupies a distinct and increasingly critical position. Unlike primary energy sources such as coal, gas, or oil, electricity represents a secondary form of energy that must be generated, transmitted, and consumed almost instantaneously. Its non-storability at scale implies that production and consumption must be continuously balanced, rendering electricity systems particularly sensitive to short-term shocks and forecast errors (Mack, 2014, p. 8). This characteristic fundamentally distinguishes electricity markets from most other commodity markets.

The European power system has undergone substantial structural change over recent decades. Driven by climate policy objectives, technological progress, and regulatory reform, electricity generation in Europe has shifted increasingly towards renewable energy sources, particularly wind and solar power. While this transition has contributed to emissions reduction and diversification of supply, it has also increased the dependence of electricity generation on weather conditions, thereby amplifying short-term variability in supply (Dannecker, 2015, pp. 2–3). At the same time, geopolitical developments and energy security concerns have reinforced the importance of well-functioning power markets and

effective price signals within Europe (Mack, 2014, p. 105; European Council, 2025; IEA, 2025, pp. 41–42).

Electricity is produced and traded through a combination of physical generation assets and organised market mechanisms. Market participants can transact electricity via centralised exchanges, such as day-ahead and intraday markets, or through bilateral over-the-counter (OTC) agreements that allow for customised contracts and longer-term hedging (IEA, 2025, p. 18; Edwards, 2023, pp. 59–60). These market structures facilitate price discovery while accommodating the diverse objectives of generators, utilities, traders, and industrial consumers. However, the continuous balancing requirement and the growing share of intermittent generation introduce additional complexity into price formation.

Electricity prices are primarily determined by supply and demand conditions, including generation capacity, fuel costs, weather-driven load patterns, and network constraints. Load refers to the real-time level of electricity consumption on the grid that must be continuously balanced by generation and therefore plays a central role in short-term price formation (Direct Energy, 2023; Edwards, 2023, p. 137). These factors are discussed in detail in Section 2.1. Beyond these structural drivers, prices are also influenced by information shocks that alter market participants' expectations. News related to weather forecasts, plant outages, grid congestion, regulatory interventions, or geopolitical developments can lead to rapid price adjustments as new information is incorporated into market valuations (Mack, 2014, pp. 105–106).

The increasing relevance of news for price formation raises the question of how textual information can be systematically incorporated into market analysis and trading decisions. Traditional quantitative models typically rely on numerical inputs and are ill-equipped to process unstructured textual data at scale. NLP offers a methodological framework to address this limitation by transforming text into structured, quantitative representations that can be analysed using statistical and machine-learning techniques. In financial applications, NLP has been employed to extract sentiment, identify themes, and quantify narratives from news and reports, thereby capturing information that is not directly observable in market data (Das, 2014, pp. 3–5; Jurafsky & Martin, 2025, pp. 4–5).

A growing body of academic literature has examined how textual information can be exploited for forecasting and trading in financial and commodity markets, providing the foundation for this present research project. Early contributions focus on news sentiment

and low-dimensional textual indicators, demonstrating that sentiment extracted from financial media contains predictive information for returns and volatility in commodity markets such as gold and oil (Smales, 2014; Li et al., 2021). These studies typically rely on aggregated sentiment measures derived from frequency-based representations and show that negative news often exerts a stronger market impact than positive news.

Subsequent research extends this line of inquiry by integrating text-derived features directly into forecasting models, treating textual information as a quantitative input alongside historical prices and fundamentals. Studies such as Guan et al. (2022) and Chakraborty et al. (2024) demonstrate that embedding-based representations of news significantly improve the prediction of commodity prices, particularly during periods characterised by geopolitical shocks or supply disruptions. This literature highlights NLP models' ability to capture expectation-driven dynamics that precede observable market movements.

More recent contributions leverage transformer-based language models and large language embeddings, enabling richer semantic representations of news content. Research by Bai et al. (2024) and Bi et al. (2025) explicitly applies these advances to electricity markets, showing that news-augmented models outperform traditional price-, load-, and weather-based benchmarks, particularly during periods of heightened volatility. These studies underline that textual information captures regulatory, infrastructural, and geopolitical developments that are not directly observable in physical system data.

Finally, a separate but related strand of literature addresses the translation of NLP-based forecasts into trading strategies, emphasising that predictive accuracy alone does not guarantee economic profitability. Studies across equity and commodity markets demonstrate that disciplined signal construction, confidence thresholds, and execution constraints are essential for transforming textual forecasts into viable trading decisions (Kirtac & Germano, 2024; J. Chen et al., 2025). This distinction between forecasting and tradable signal generation is particularly relevant in electricity markets, where price dynamics are fast, volatile, and subject to institutional constraints.

Against this background, the present research project positions itself at the intersection of electricity market analysis and modern text analytics. By investigating how news-based information can be transformed into quantitative signals using NLP techniques, the research project seeks to deepen understanding of information processing in power markets

and to explore the potential value of textual data for electricity price forecasting and trading strategies.

## 1.2 Research Question

This research project examines the following research question:

*Can energy-related news be transformed into statistically and economically significant trading signals for the German power market using a dual Natural Language Processing framework based on transformer models and semantic embeddings?*

To address this question, the study employs a structured modelling approach that combines quantitative market information with textual data. The baseline feature set comprises not only electricity price series but also broader power market indicators, such as baseload prices and temporal variables capturing intraday, weekly, and seasonal patterns. These features are augmented with NLP-based indicators extracted from energy-related news headlines, which are transformed into quantitative representations using transformer-based language model

s and embeddings. The integration of these feature sets enables the evaluation of whether news-derived information provides incremental predictive and economic value beyond traditional market-based inputs. Furthermore, the economic significance of these signals is assessed through a comprehensive backtesting framework, which evaluates the performance of the news-augmented signal model against both the baseline model and a naïve long-only strategy. The detailed data construction, model architecture, and evaluation framework are presented in Chapter 3.

## 1.3 Objective and Significance

Electricity prices are highly sensitive to information that affects expectations about short-term supply and demand, including news on renewable generation, outages, weather conditions, policy interventions, and input costs such as natural gas, coal, and $CO_2$ allowances. While news-based forecasting and sentiment-driven trading strategies are well established in equity and commodity markets, comparatively little academic research has examined how electricity market news can be transformed into statistically and

economically meaningful trading signals. This gap is notable given the strong role of expectations and information shocks in short-term power markets.

The objective of this research project is to assess whether systematically incorporating energy-related news into electricity market models provides incremental value relative to approaches based solely on historical prices and temporal patterns. In particular, the study evaluates whether news-informed models improve both predictive accuracy and economic performance when applied to short-term trading decisions in the German power market.

The study contributes to the literature by extending recent advances in NLP and embedding-based text analysis to electricity markets and by explicitly distinguishing between forecasting performance and tradable signal construction. From a practical perspective, the results provide insight into whether structured processing of news can enhance short-term decision-making in power markets, where rapid information assimilation and disciplined execution are essential.

## 1.4 Structure of the Research Project

This research project is structured as follows. Chapter 2 presents the theoretical foundations and related literature, covering the fundamentals of electricity markets, key drivers of price formation, and the conceptual foundations of NLP and text-based trading signal generation. This chapter establishes the economic and methodological context for the empirical analysis. Chapter 3 outlines the data sources, modelling framework, and methodological pipeline employed in the study. It describes the construction of market-based and news-derived features, the NLP models used to process textual information, and the procedures for model training, validation, and evaluation. Chapter 4 presents the empirical results of the forecasting and trading analyses. Model performance is evaluated using both statistical metrics and economic criteria, and the result of the news-augmented signal model is compared against the baseline benchmarks without news information and a naïve long-only strategy. Chapter 5 concludes the research project by summarising the main findings, discussing limitations, and outlining implications for future research and practical applications in electricity markets. Chapter 6 provides a conclusion to the entire research project.

# 2 Theoretical Foundations

This chapter reviews academic literature and theoretical foundations of electricity price formation, news-based forecasting and the application of NLP in financial and energy markets. The analysis begins in Section 2.1, which provides a general overview of the power market, its institutional structure, and the fundamental drivers of electricity prices. Section 2.2 then outlines the fundamental concepts and methodological foundations of NLP, while Section 2.3 surveys extant research on news-based forecasting and trading applications in financial and commodity markets. Together, these sections establish the conceptual foundation for the empirical methodology developed in subsequent chapters.

## 2.1 Fundamentals of the Electricity Market

This section examines the structural and economic characteristics of electricity markets relevant to short-term price formation. The text provides an overview of the unique physical properties of power (Section 2.1.1), the roles of key market participants (Section 2.1.2), and the mechanisms of power trading (Section 2.1.3). Moreover, the section undertakes an analysis of the fundamental drivers of supply and demand (Section 2.1.4) and the subsequent trading behaviours (Section 2.1.5)

### 2.1.1 Overview of Electricity Market Structures

The evolution of electricity systems has been characterised by the development of energy sources, with periods of both stagnation and rapid advancement. In recent times, there has been a resurgence in the adoption of renewable energy sources, marking a modern renaissance in this field. The 1970s oil crisis led to accelerated policy and technological shifts, which resulted in the revival of non-fossil energy sources. Today's power systems increasingly rely on a diverse mix of energy sources like wind and solar power, along with established hydropower (Lefort & Simic, 2024). The underlying principle of renewable sources is that they are unlimited and not finite, in contrast to fossil fuels, which are utilised for conventional energy generation (Bundesnetzagentur, 2025a). As demonstrated by Statista (2025), the German market has experienced a notable increase in renewable electricity production, with wind power witnessing a 36% growth and solar power recording a substantial 64% increase from 2018 to 2025. This expansion plays an important role

in reducing emissions from electricity generation. However, as Dannecker (2015, p. 17) illustrates in the German context, it also brings certain challenges and unintended side effects. Wind and solar energy production is highly dependent on weather conditions, resulting in inconsistent output and impacting electricity prices, which leads to increased volatility. Countries that want to rely heavily on renewables, such as Germany, are particularly affected by these fluctuating energy prices. A more recent example of this phenomenon is the recent blackout that occurred in Spain this year, which was attributed to an excess of renewable energy generation and a deficit of conventional energy (CNN, 2025). In this context, Dannecker (2015, p. 20) highlights the need for more advanced forecasting methods for renewable electricity, ensuring that production can be better aligned with demand and helping to stabilise the increased volatility observed in modern power markets.

### 2.1.2 Participants in Electricity Markets

Electricity generated at power plants is fed into the high-voltage transmission grid, where it can be transported over long distances with minimal losses. Transmission system operators (TSOs) manage this high-voltage network by ensuring that electricity from various generation facilities is safely integrated, routed, and delivered to regional distribution grids. At the European level, TSOs are coordinated by European Network of Transmission System Operators for Electricity (ENTSO-E), which harmonises grid operation, security standards, and cross-border electricity flows to support a reliable and interconnected power system (Bundesnetzagentur, 2025c). The transmission grid is the fundamental component of the electricity supply system, while the distribution grid delivers electricity to households, firms, and small industrial consumers. Market participants purchase electricity on exchanges such as the European Power Exchange (EPEX) to either supply customers, hedge their exposures, or optimise their own generation portfolios (Bundesnetzagentur, 2025b). A notable issue is the relative immaturity of electricity as a commodity, in contrast to well-established products such as gold, coffee, and oil, which are easily stored. Electricity, by contrast, must be consumed immediately, creating distinct challenges for its storage, pricing, and distribution (Edwards, 2023, p. 2).

### 2.1.3 Power Trading Instruments

Power producers agree to sell electricity to buyers at pre-agreed terms via purchase agreements or long-term contracts, often structured physically or financially. These instruments are used to secure price certainty, facilitating financing, or, in some cases, to obtain exposure to market price fluctuations for speculative purposes (Edwards, 2023, p. 2). This can be achieved through bilateral OTC contracts or by trading through an intermediary such as EPEX, where transactions can be executed in the spot market or via derivatives (Edwards, 2023, p. 5). The present research project focuses on spot and day-ahead trading, while other contract forms are included for the sake of completeness, but fall outside the study's scope.

The two electricity markets under discussion in this research paper comprise the day-ahead auction and the intraday market, where they enable short-term balancing of supply and demand. The intraday market allows continuous trading, enabling participants to adjust positions in response to updated forecasts or unexpected changes in generation and consumption (EPEX, 2025). A distinctive feature of the spot market is that purchased electricity must be consumed immediately, which requires direct physical delivery to the end user. As a result, the spot market is accessible primarily to companies that are able to either take delivery or utilise the electricity directly. It is important to recognise that the spot market is characterised by pronounced volatility, driven by a combination of internal and exogenous factors. Unexpected events, such as unplanned outages, can lead to sharp price increases as the market reacts to sudden shifts in supply (Edwards, 2023, p. 51). Conversely, electricity prices may also become negative – an uncommon occurrence in commodity markets – when excess generation exceeds system demand and the system's ability to absorb surplus through flexible demand, storage, curtailment, or cross-border flows. Negative prices therefore function as a market signal to curb inflexible generation and, where possible, stimulate additional consumption, helping the system absorb temporary oversupply arising from high renewable feed-in and low demand (EPEX, 2025).

Building on the characteristics of the spot market where prices can fluctuate sharply due to volatility, scarcity, or even negative pricing, it is also important to consider how market participants manage these dynamics through longer-term instruments. In the day-ahead auction, bids and offers for each hourly delivery period of the following day are aggregated into a uniform market-clearing price once the auction closes at 12:00 CET (EPEX, 2025). Beyond these hourly products, electricity is also traded using standardised delivery

profiles. A base load contract represents all 24 hours of a given delivery period – such as a day, week, month, or year – and its price corresponds to the average of all hourly prices within that interval. A peak load contract, by contrast, covers only the hours with the highest typical demand, usually from 08:00 to 20:00 on working days, with its price reflecting the average of the hourly prices during this peak window, which are generally higher due to increased consumption (Müsgens & Bade, 2024, pp. 15–16).

Beyond these standardised delivery profiles, electricity markets also employ longer-term contractual instruments that allow price risks to be managed across extended horizons. Forward contracts are typically negotiated bilaterally OTC and enable buyers and sellers to fix prices for future delivery periods, offering a high degree of flexibility with respect to maturity, volume and settlement structure (IEA, 2025, p. 62). Closely related to forwards, electricity swaps represent financially settled agreements that exchange fixed and floating price components and are widely used for risk management purposes. Complementing these bilateral instruments, futures contracts provide a standardised alternative that is traded on exchanges and cleared centrally, implying daily margining and reduced counterparty risk (IEA, 2025, p. 62; Mack, 2014, p. 52). Options further extend the set of available instruments by granting the right, but not the obligation, to buy or sell electricity or related underlyings, thereby allowing market participants to limit downside risk while retaining exposure to favourable price movements.

The coexistence of different delivery profiles, such as base and peak load, together with longer-term contractual instruments highlights the need to accommodate fluctuating electricity consumption and evolving risk exposures across time. While delivery profile's structure when electricity is supplied, forward and derivative products enable price risks associated with these profiles to be managed over extended horizons. Taken together, these instruments provide the foundation for organising trading activity in electricity markets and for translating physical system characteristics into market prices. Against this background, it becomes essential to consider the internal and external factors that influence price formation and shape the functioning of electricity markets more broadly.

### 2.1.4 Supply and Demand Dynamics in Electricity Markets

The existence of different delivery profiles, such as base and peak load, alongside the products traded in the spot and day-ahead markets, reflects the fact that electricity

consumption does not remain constant throughout the day. These products allow market participants to address varying usage patterns and to handle the price movements that result from them. Against this background, it becomes essential to consider the supply and demand factors that shape price formation and determine how the electricity market functions more broadly. Although these factors are often conceptually separated, they are closely interlinked in practice, which means that a strict distinction is not always possible.

Previous sections have already highlighted demand-related factors such as base and peak load, which reflect the fact that electricity usage varies across the hours of the day. In addition, weekly cycles can also be observed, with noticeable differences between weekdays and weekends. These variations are largely driven by the recurring consumption behaviour of households and firms. Generators attempt to align production with these demand profiles as closely as possible, relying on forecasting models that are calibrated to expected usage patterns (Edwards, 2023, p. 135). Beyond daily and weekly patterns, electricity markets are also shaped by broader seasonal effects. The contrast between summer and winter plays a more pronounced role than in markets for storable commodities. For many of these commodities, storage creates a convenience yield – an economic benefit of holding inventories – that often contributes to a smoother, upward-sloping term structure. Many energy markets, including gas and electricity, by contrast, cannot rely on economically viable large-scale storage, and their prices therefore tend to exhibit more distinct seasonal patterns, with higher levels in winter and lower levels in summer due to the corresponding shifts in demand (Edwards, 2023, p. 77). Edwards (2023, p. 137) further shows a strong correlation between temperature and load. While electricity demand increases in winter due to heating requirements, hot summer days can likewise result in elevated consumption as cooling systems operate more intensively, thereby increasing the need for additional electricity supply.

Beyond these structural demand patterns, electricity markets are also influenced by broader macroeconomic conditions. Economic growth, industrial activity and business cycles affect electricity consumption through changes in production levels, investment activity and household demand, linking electricity prices to the general macroeconomic environment. Periods of economic expansion tend to raise electricity demand and tighten capacity margins, while economic slowdowns reduce load growth and ease price pressure (Edwards, 2023, pp. 51 & 114).

As with demand, the supply side of the electricity market is shaped by a combination of technological, operational, meteorological and regulatory factors, which together determine both the maximum available generation and the marginal cost at which it can be supplied.

A central supply-side driver is the cost of fuels and other inputs, since these determine the marginal cost of conventional generation and thus the order in which power plants are dispatched. In economic terms, the relevant benchmark is often the levelized cost of electricity, which aggregates initial investment, fuel, operation and maintenance, and financing costs into a single unit cost over the lifetime of a plant (Mack, 2014, p. 212). For fossil-fuel technologies such as coal and gas, a large share of this cost is driven by fuel prices, so changes in coal, gas or $CO_2$ allowance prices feed directly into the short-run marginal cost of generation (Edwards, 2023, p. 138). By contrast, wind and solar are characterised by high upfront investment to build the plant but very low marginal costs, as additional output does not require extra fuel buying (IEA, 2025, pp. 24–25). This difference in cost structure explains why, once built, renewable plants typically enter the merit order before conventional units and push higher-cost generators to the margin. The merit order is a ranking of available power plants from lowest to highest marginal cost, ensuring that the cheapest energy sources are always used first to meet electricity demand. (Müsgens & Bade, 2024, pp. 6 & 96; Edwards, 2023, p. 131).

Beyond cost considerations, the technical characteristics of generating units play an important role in determining available supply. Conventional plants differ in their ramp-up and ramp-down capabilities, minimum stable generation levels and start-up times, which constrain their ability to respond quickly to variations in demand or renewable production (Edwards, 2023, p. 28). Intermittent renewable sources such as wind and solar exhibit strong weather-related fluctuations and relatively low average availability compared with their installed capacity. A study of hourly load factor analyses in Germany reveals that wind power plants achieve a load factor of 60% during a mere 5% of the year. When considering the full distribution of results, there is a 90% probability that the hourly load factor remains between 4% and 55%. The wide range and high frequency of low-output periods confirm that only a limited fraction of the total installed fleet can be considered reliably available for system planning at all times (Dannecker, 2015, pp. 18–19). Edwards (2023, p. 358) likewise notes that wind output is inherently variable and sensitive to seasonal, geographical and day time conditions. In addition to these technical and

meteorological constraints, not all installed capacity is continuously usable. Parts of the generation fleet are periodically unavailable due to scheduled maintenance or unexpected outages. For this reason, adequacy assessments distinguish between unavailable and reliably available capacity, with system operators also maintaining reserve capacity to manage forecast errors and short-term imbalances (Dannecker, 2015, p. 19).

An additional factor that influences supply is the physical capacity of the transmission grid, which determines the maximum amount of electricity that can reach consumers. Even when sufficient power is produced, congestion in the high-voltage network can limit transfers between regions, requiring TSOs to intervene through redispatch measures or even grid shutdowns to maintain system stability and prevent overloads (Dannecker, 2015, p. 23). In such situations, the physical limits of the grid – or transmission congestion – lead to regional price differences, often determined by Locational Marginal Pricing (LMP), as constrained areas experience price variations depending on local supply and demand conditions. LMP is a pricing method where the price of electricity is calculated at different points across the transmission grid, instead of using one single price for different regions (Edwards, 2023, p. 133; Müsgens & Bade, 2024, p. 107). As this study focuses on price forecasting within a single market, the detailed modelling of cross-border dynamics will not be examined.

Given these structural, technical, and market-driven dynamics, the functioning of electricity markets is increasingly shaped by the regulatory frameworks that govern investment incentives, system reliability, and market behaviour. Policy instruments are essential to stabilize operations and maintain security of supply amid growing renewable penetration and heightened price volatility (IEA, 2025, pp. 7–9). Regulators have expanded the use of mechanisms such as capacity remuneration schemes to pay generators for their available capacity, ensuring sufficient dispatchable power remains available as variable renewables grow (IEA, 2025, p. 84). Simultaneously, authorities are tightening transparency and monitoring obligations to mitigate risks linked to market manipulation and algorithmic bidding, thereby preserving market integrity in short-term markets (IEA, 2025, p. 29; Müsgens & Bade, 2024, p. 173). Current policy discussions emphasise the need for market designs and regulatory tools to align, ensuring affordability, investment certainty, and long-term system reliability (IEA, 2025, p. 15).

Beyond domestic regulation, geopolitical factors and international energy policies play an important role in shaping electricity market conditions. Decisions by major energy-producing countries and organisations, such as OPEC, influence global oil and gas prices, which indirectly affect electricity generation costs in fuel-dependent systems through price transmission mechanisms (Mack, 2014, p. 182; Edwards, 2023, p. 162). In European power systems, fossil fuel prices are particularly relevant because thermal generation technologies are often positioned at the margin of the merit order and therefore contribute directly to short-run price formation when renewable output is insufficient (Edwards, 2023, pp. 28–29). Geopolitical developments, including supply disruptions, sanctions, or changes in export policy, can thus trigger abrupt revisions in fuel price expectations and amplify price volatility in electricity markets, even though their effects materialise indirectly through input costs rather than through electricity demand itself (IEA, 2025, p. 7; Mack, 2014, p. xx).

Taken together, electricity price formation reflects the interaction of demand patterns, generation costs, network constraints, regulatory frameworks and external macroeconomic and geopolitical influences. These factors do not operate in isolation but interact in a dynamic and sometimes non-linear manner, contributing to the pronounced volatility observed in electricity markets. Recognising this complex risk environment is essential for analysing electricity prices and provides the foundation for the forecasting approaches examined in the subsequent sections.

### 2.1.5 Motivation and Established Approaches to Power Trading

Müsgens and Bade (2024) and Edwards (2023) have demonstrated that electricity market instruments exhibit strong parallels to those used in traditional financial markets. While the focus of this study is on spot and day-ahead data, the market structure is also comprised of products such as forwards, futures, and other derivatives. Consequently, this section will highlight these other products to provide a comprehensive introduction but will focus primarily on the short-term power markets.

As established in Section 2.1.3, the day-ahead and intraday markets form the core of short-term electricity trading. The current section analyses the trading behaviours and motivations of market participants within these venues. The inherent properties of electricity – its non-storability, high volatility, and sensitivity to real-time information – are

the fundamental drivers shaping the distinct and specialised trading strategies employed by market participants (Edwards, 2023, pp. 49–50; Müsgens & Bade, 2024, p. 23).

Motivations differ substantially across actor groups. Utilities and large industrial consumers primarily use short-term markets for hedging, securing expected consumption at predictable prices and thereby reducing exposure to fluctuations in spot prices and standard load profiles, which represent the average quarter-hourly electricity consumption of typical households (Müsgens & Bade, 2024, pp. 39–40). To complement these short-term hedges, many large consumers enter into forward contracts, which allow them to stabilise procurement costs over longer horizons by fixing part of their expected load at predetermined prices (IEA, 2025, pp. 62–63). Generation companies align their trading activity with operational optimisation. They monitor indicators such as clean spark and dark spreads, which compare electricity prices with fuel and $CO_2$ costs, to assess whether operating a power plant yields a positive margin. These spreads reflect the basic economics of thermal generation, where operators pay for fuel and $CO_2$ certificates while receiving revenue from electricity sales, making the spread a direct measure of short-term profitability. These signals inform dispatch decisions and influence whether conventional units are ramped up, reduced or temporarily shut down. Because these spreads replicate the marginal profit of specific generation technologies, they are also traded OTC as financial products, allowing market participants to hedge or speculate on relative movements between power, fuel and emission prices (Müsgens & Bade, 2024, pp. 66–67; Edwards, 2023, p. 79).

In addition to spot and forward markets, many actors rely on exchange-traded futures and options, which offer liquidity, transparency and central clearing. Utilities and retailers predominantly use futures to manage portfolio risk in a more flexible manner than with bilateral forwards, as futures allow frequent rebalancing through daily margining and can be unwound at low transaction costs in liquid trading sessions (Mack, 2014, pp. 133–134; IEA, 2025, pp. 62–63). Generators employ futures to hedge aggregated production volumes over broader periods rather than individual delivery windows, benefiting from standardised contract sizes that simplify portfolio hedging across multiple months or seasons (Mack, 2014, pp. 141–142). Options are used when actors seek asymmetric protection: market participants buy call or put options to insure against extreme price movements, while option writers earn premiums when expecting stable market conditions (Mack, 2014, p. 143). Derivative markets therefore enable both hedgers and speculators

to manage or assume risk efficiently while contributing to market liquidity (Mack, 2014, p. 167).

Other speculators in electricity markets include proprietary traders, who engage solely for financial returns without any link to physical delivery (Müsgens & Bade, 2024, p. 35), as well as power plant operators, who may pursue speculative opportunities in addition to their standard activity of producing and selling electricity. Proprietary traders and speculative desks adopt opportunity-driven strategies that respond to short-term market fluctuations. Such fluctuations often arise from unexpected forecast revisions or rapid changes in renewable production, which can create temporary price distortions in intraday spot markets (Müsgens & Bade, 2024, pp. 88–90). Their objective is to generate profit from the inherent price volatility in electricity markets, where even slight discrepancies between real-time generation and consumption can prompt price fluctuations, which in turn can be used as trading opportunities (Edwards, 2023, pp. 51 & 241).

The growing operational complexity of electricity systems has also encouraged the use of algorithmic order execution within the intraday market. Automated systems monitor updated weather forecasts, TSO data and plant availability, enabling rapid adjustments to intraday positions and helping to minimise balancing costs for portfolio managers. Such algorithms are often used to close deviations in balancing groups immediately following forecast changes. Nevertheless, discretionary trader judgment remains essential when market conditions deviate from typical patterns or when qualitative information, such as unexpected outage announcements or operational restrictions, must be interpreted (Müsgens & Bade, 2024, pp. 153–154).

Given these behavioural and technological aspects, trading outcomes are further shaped by the risk environment characteristic of electricity markets, as outlined in Section 2.1.4. High intraday volatility, frequent price spikes and uneven liquidity across products require disciplined management of mark-to-market exposures and position limits (Mack, 2014, p. 8). As Mack (2014, p. 8) notes, traders must account not only for price risk but also for operational uncertainties arising from outages and transmission congestion, as these events can rapidly alter expected market conditions. The need to react to such risks in real time places considerable emphasis on timely information processing and on the ability to adjust trading strategies swiftly in response to changing fundamentals.

Taken together, these trading practices show that electricity markets depend heavily on rapid information processing, continuous position management and flexible optimisation. They also provide a strong motivation for incorporating news-based and machine-learning approaches into price forecasting, as examined in the following sections.

## 2.2 Fundamentals of Natural Language Processing

This section provides a methodological foundation for the construction of feature-based news content. The text introduces the role of NLP in analysing unstructured text (Section 2.2.1), the transformation of text into quantitative embeddings (Section 2.2.2), and the specific challenges of applying these methods within financial contexts (Section 2.2.3).

### 2.2.1 Introduction to Natural Language Processing

NLP is a subfield of machine learning that studies how computers can be used to analyse, interpret and generate human language in textual or spoken form (Lee, 2023, pp. 3–4). In contrast to structured numerical data, natural language is inherently unstructured, ambiguous and context dependent. The meaning of linguistic expressions depends not only on individual words but also on grammatical structure, surrounding context and domain-specific conventions, which makes automated processing particularly challenging (Lee, 2023, p. 5).

In economic and financial research, NLP has gained importance due to the increasing availability of large volumes of textual information, including news articles, policy statements and corporate disclosures, which contain information that is not immediately reflected in quantitative market data (Das, 2014, pp. 3–4). Financial markets are influenced not only by realised economic outcomes but also by expectations, uncertainty and narratives communicated through language. NLP provides a systematised framework for the conversion of unstructured textual information into quantitative representations. These representations can then be integrated into empirical models and decision-making processes, which can be employed for the generation of trading signals (Das, 2014, pp. 4–5).

Earlier approaches relied primarily on frequency-based representations, whereas modern NLP increasingly applies neural language models that can capture contextual meaning and semantic relationships more effectively (Jurafsky & Martin, 2025, p. 4). Recent

progress in representation learning, particularly through transformer-based architectures, has substantially improved the ability of NLP models to account for context-dependent semantics (Jurafsky & Martin, 2025, p. 171). These models produce dense numerical representations, commonly referred to as embeddings, which encode both lexical meaning and contextual relationships within a text. Such advances have expanded the applicability of NLP to information-sensitive domains such as finance and energy markets, where small changes in wording may signal shifts in expectations or market conditions (Lee, 2023, p. 118; Das, 2014, p. 4; Simonian, 2025, p. 127).

### 2.2.2 Conceptual Foundations of Text-to-Quantitative Transformation

A fundamental task in NLP is the transformation of unstructured textual data into numerical representations that can be analysed using statistical and machine-learning methods. Unlike numerical variables, text does not possess an inherent quantitative structure, which requires explicit encoding schemes that preserve linguistic meaning while enabling mathematical processing (Jurafsky & Martin, 2025, p. 4). The objective of text representation is therefore to map words, sentences or documents into vector spaces in which semantic similarity can be expressed through numerical distance (Jurafsky & Martin, 2025, p. 99).

Early approaches to text representation rely on frequency-based techniques, such as bag-of-words and term-frequency measures, which describe documents by counting word occurrences without considering word order or context (Jurafsky & Martin, 2025, pp. 4 & 235). To improve their descriptive power, term weighting schemes such as Term Frequency–Inverse Document Frequency (TF-IDF) assign higher importance to words that are frequent within a document but rare across the corpus, thereby reducing the influence of common yet uninformative terms (Jurafsky & Martin, 2025, pp. 235–236; Simonian, 2025, p. 129; Das, 2014, pp. 13 & 81). While these methods are transparent and computationally simple, they fail to capture semantic relationships between words and texts. More advanced representation methods address this limitation by embedding textual elements into continuous vector spaces based on their contextual usage, thereby allowing semantically related expressions to be represented more closely (Jurafsky & Martin, 2025, p. 95; Lee, 2023, p. 118). Embedding-based representations therefore outperform frequency-based approaches because they capture meaning through contextual co-occurrence patterns rather than isolated word counts. Words or texts that appear in similar

contexts are mapped to nearby positions in the vector space, even if they do not share identical vocabulary (Jurafsky & Martin, 2025, pp. 4 & 101). In practice, semantic similarity is encoded through geometric relationships, where cosine similarity or Euclidean distance between vectors provides a quantitative measure of conceptual relatedness. While Euclidean distance calculates the absolute spatial magnitude between two points, cosine similarity evaluates the angular orientation of vectors, offering a robust metric that remains independent of document length or vector magnitude (Lee, 2023, pp. 326–327).

While embedding-based representations capture semantic similarity through vector geometry, early embedding models assign a single representation to each word, regardless of its linguistic context. Transformer-based language models extend this framework by producing contextual embeddings, where word representations are dynamically adjusted based on surrounding tokens within a sentence (Jurafsky & Martin, 2025, p. 171). This architecture enables the model to learn complex pairwise relationships and long-range dependencies, producing contextual embeddings that adjust dynamically to a word's usage. These dynamic representations encode rich semantic and syntactic information that can be directly applied to classification and topic modelling. By addressing challenges such as polysemy, where a single word can have multiple meanings, transformers are particularly effective in financial applications. They allow for the detection of subtle contextual cues – such as shifts in tone, uncertainty, or narrative framing – that often precede observable market movements (Jurafsky & Martin, 2025, pp. 172 & 206; Das, 2014, p. 4).

While text representation converts language into numerical form, the analysis of this data typically follows two distinct paradigms: classification and topic modelling. Classification is a supervised learning task where documents are assigned to predefined categories based on labelled training data. Typical applications include sentiment classification, event detection or relevance filtering, where the goal is to map textual inputs to known categories (Lee, 2023, pp. 336–337; Jurafsky & Martin, 2025, pp. 61–62). In contrast, topic modelling is an unsupervised approach that identifies latent thematic structures within a corpus without prior labelling. Rather than assigning texts to explicit categories, topic models identify clusters of semantically related terms that co-occur frequently across documents, thereby revealing dominant themes present in the data (Lee, 2023, p. 125; Das, 2014, p. 42). In financial research, these techniques enable the systematic transformation of qualitative narratives, including news flows and policy communications,

into structured signals that can be linked to quantitative outcomes, making them particularly valuable for discovering emerging risks that have not yet been captured by traditional market indices (Loughran & McDonald, 2020, p. 2; Simonian, 2025, p. 127).

The high dimensionality of modern text embeddings introduces computational and statistical challenges for both classification and topic modelling. Contextual embeddings generated by transformer models typically inhabit high-dimensional vector spaces, which can impede the efficiency and accuracy of clustering algorithms due to the "curse of dimensionality" (Grootendorst, 2022, p. 2). To address this, dimensionality reduction techniques are employed to project these embeddings into a lower-dimensional space while preserving their essential topological structure. The Uniform Manifold Approximation and Projection (UMAP) algorithm is frequently preferred for this task, as it effectively maintains both local and global semantic relationships, ensuring that semantically similar documents remain grouped together in the reduced space (McInnes et al., 2020, pp. 1–2).

Once embeddings are represented in a reduced space, clustering algorithms can be applied to identify groups of semantically related texts. Building on the unsupervised paradigm previously noted, this process is especially relevant for uncovering latent structures in large, heterogeneous document collections (Das, 2014, p. 42). Density-based approaches such as HDBSCAN are well suited for this purpose, as they can identify clusters of varying size and density while explicitly modelling noise, a common feature of real-world textual datasets (Grootendorst, 2022, p. 3; McInnes & Healy, 2017, pp. 1–2). Together, dimensionality reduction and clustering form a critical conceptual bridge between high-dimensional embeddings and interpretable topic structures in NLP applications.

The final step in the transformation process is the extraction of interpretable labels from the identified clusters. While standard TF-IDF evaluates word importance at the document level, the class-based TF-IDF (c-TF-IDF) procedure treats all documents within a cluster as a single class. This allows the model to calculate a score that reflects how representative a term is of a specific topic by comparing its frequency within the cluster to its frequency across the entire corpus. By identifying words that are unique to specific clusters, c-TF-IDF provides a precise linguistic description of the underlying market narratives or economic themes captured by the model (Grootendorst, 2022, pp. 3–4).

The conceptual process described in this section is implemented using Bidirectional Encoder Representations from Transformers (BERT), a transformer-based language model

that jointly performs tokenisation, contextual embedding generation and semantic representation (Devlin et al., 2019, pp. 1–2; Jurafsky & Martin, 2025, p. 171). BERT encodes linguistic meaning as dense vectors that capture both semantic and syntactic relationships through contextualised representations rather than fixed word-level statistics (Jurafsky & Martin, 2025, p. 197).

This research paper employs specialised variants of the BERT architecture that are tailored to the computational and analytical requirements of large-scale text analysis. Compact models such as MiniLM reduce computational complexity while preserving much of the semantic representation capacity, making them suitable for large-scale text analysis (Wang et al., 2020, p. 1). In addition, zero-shot classification models based on transformer embeddings enable semantic categorisation without task-specific labelled data, relying on general language understanding acquired during pretraining (Yin et al., 2019, pp. 1–2). The specific implementation choices and empirical integration of MiniLM and zero-shot are presented in detail in Chapter 3.

### 2.2.3 Limitations and Challenges of NLP in Financial Applications

Despite its growing relevance, the application of NLP in financial contexts faces several methodological and practical limitations. A central challenge arises from the ambiguity and context sensitivity of financial language, where identical terms may carry different meanings depending on macroeconomic conditions, institutional settings or market regimes. This linguistic complexity increases the risk of misinterpretation, particularly when models are applied across heterogeneous sources such as news articles, policy statements and analyst commentary (Das, 2014, pp. 105–107).

Another limitation concerns data quality and representativeness. Financial text data often contain noise, redundancy and reporting bias, as media coverage tends to concentrate on extreme events or prominent institutions. Such distortions can lead NLP models to overemphasise sensational narratives while underrepresenting gradual or structural developments, potentially weakening the stability of derived signals (Loughran & McDonald, 2020, p. 3; Das, 2014, p. 105). Furthermore, changes in language use over time, driven by regulatory shifts or evolving market discourse, may reduce model performance if not adequately addressed through retraining or adaptive techniques (Loughran & McDonald, 2020, pp. 4–5).

From a modelling perspective, transformer-based architectures introduce challenges related to interpretability and computational cost. While contextual embeddings offer superior semantic representation, their internal decision processes remain largely opaque, complicating economic interpretation and model validation (Wang et al., 2020, p. 1; Das, 2014, p. 107; Jurafsky & Martin, 2025, p. 172). In addition, the reliance on large pre-trained models raises concerns regarding domain mismatch, as models trained on general language corpora may not fully capture specialised financial or energy-market terminology without further adaptation (Kaplan et al., 2023, p. 1; Das, 2014, p. 105; Jurafsky & Martin, 2025, p. 162).

A further methodological limitation arises from the use of non-linear dimensionality reduction techniques such as UMAP. While it is widely applied, the algorithm relies on stochastic optimisation procedures and an explicit random initialisation seed (McInnes et al., 2020, pp. 1–4). As a result, repeated runs on identical data may yield slightly different low-dimensional representations, even when model parameters are held constant. Although these variations are typically small, they can be amplified in downstream tasks such as clustering or topic extraction if not carefully controlled. In empirical applications, this non-determinism introduces an additional source of variance that must be managed through fixed random seeds, robustness checks, or repeated sampling, particularly when dimensionality-reduced embeddings are used as inputs to subsequent modelling stages rather than for exploratory visualisation alone (Grootendorst, 2022, pp. 2–4).

Taken together, these limitations highlight the importance of careful model selection, robust preprocessing and complementary quantitative validation when deploying NLP techniques in financial and energy market research.

## 2.3 Fundamentals of Trade Signal Generation Using NLP

Building upon the linguistic foundations of Section 2.2, this section examines the practical application of NLP within financial and energy markets. The text distinguishes between two concepts: statistical forecasting (Section 2.3.1) and actionable trading implementation (Section 2.3.2). The former focuses on demonstrating the predictive power of news beyond quantitative inputs, while the latter addresses the transformation of these forecasts into directional signals using threshold rules and risk controls. Attention is given to empirical applications in energy and electricity markets, where rapid information

processing and expectation-driven price dynamics make NLP-based trading strategies especially relevant.

### 2.3.1 From Textual Information to Market Forecasts

Building on the power market characteristics discussed in Section 2.1 and the text-to-vector transformation outlined in Section 2.2, recent research integrates textual information into forecasting models for energy and commodity prices. Standard forecasting setups typically rely on historical prices, load patterns, weather variables, and physical system constraints, which capture structural and seasonal behaviour but only partially reflect forward-looking information (Hong et al., 2020, pp. 377–378; Müsgens & Bade, 2024, pp. 5–6). Textual sources such as news and policy communication can contain expectation- and uncertainty-related signals that precede observable price adjustments, making them a natural complement to purely quantitative inputs (Das, 2014, p. 4; Loughran & McDonald, 2020, pp. 1–2). In energy markets, this channel is particularly relevant because regulatory and geopolitical developments may shift price expectations before physical effects materialise (Dannecker, 2015, pp. 2–3; Edwards, 2023, p. 162).

Early empirical studies linking news to market outcomes focused on sentiment indicators derived from textual data. Lee Smales (2014) examines the gold futures market using commodity-specific news sentiment scores obtained from structured news analytics. The study aggregates sentiment indicators extracted from financial media and incorporates them into regression-based forecasting models alongside historical price information news (Smales, 2014, pp. 277–278). Using daily gold futures data over an extended sample period, Smales (2014, pp. 285–286) shows that news sentiment significantly improves the prediction of short-term returns and volatility, with negative news exerting a stronger market impact than positive news.

Similarly, Li et al. (2021) analyse oil futures markets and construct sentiment measures from oil-related news articles. Their framework combines textual sentiment indicators with traditional econometric volatility models, demonstrating that sentiment contains incremental predictive information for both returns and volatility, particularly during periods of elevated uncertainty (Li et al., 2021, pp. 3–4). These early contributions treat sentiment as a low-dimensional numerical signal derived from text, consistent with the frequency-based and term-weighting approaches discussed in Section 2.2.2.

Subsequent research extends this idea by embedding textual features directly into forecasting models rather than relying solely on aggregated sentiment scores. Guan et al. (2022) propose a forecasting framework for international natural gas prices that combines historical price series with numerical representations of related news articles. Textual information is first transformed into quantitative features and then integrated into machine-learning models alongside lagged price inputs (Guan et al., 2022, pp. 5–6). Their empirical results show that news-augmented models outperform price-only benchmarks, especially in capturing abrupt price movements associated with geopolitical events and supply disruptions (Guan et al., 2022, p. 10). This approach illustrates how text embeddings, as introduced in Section 2.2.2, can be treated as additional explanatory variables within standard forecasting pipelines.

More recent studies leverage advances in transformer-based language models to extract richer semantic representations from news. Chakraborty et al. (2024, pp. 3–4) develop an event-driven forecasting framework in which news articles are encoded using contextual embeddings derived from transformer models. Their Robust Event Network (REN) model is a neural network-based predictive model that specifically utilises Long Short-Term Memory hidden states to capture non-linear, long-term dependencies between extracted event intensities and commodity price fluctuations (Chakraborty et al., 2024, pp. 6 & 13). Rather than modelling sentiment alone, this framework explicitly links semantic content to price reactions. This aligns with the argument of Kaplan et al. (2023) that general language models require asset-specific adaptation to capture economically relevant information in energy and commodity markets (Kaplan et al., 2023, p. 1). The authors demonstrate that incorporating exogenous event features from news significantly improves accuracy for commodities sensitive to real-world developments. They highlight that while traditional models often fail to account for extreme volatility, their approach captures the impact of external shocks and provides early warning signals for abrupt price fluctuations following significant information releases (Chakraborty et al., 2024, pp. 24–25).

While RENs rely on deep learning architectures, other high-performing frameworks in financial forecasting frequently employ gradient-boosted decision trees to integrate these textual embeddings with numerical data. Methods such as XGBoost are particularly effective in this context as they iteratively combine multiple weak learners into a strong predictive model through additive training, minimizing forecast errors while efficiently

capturing complex, non-linear interactions (T. Chen & Guestrin, 2016, pp. 1–2; Simonian, 2025, pp. 54 & 65).

Bai et al. (2024, pp. 1–2) investigate the joint role of news and load information in electricity price forecasting. Their framework combines traditional physical variables, such as demand and load forecasts, with textual features extracted from energy-related news. Using supervised learning models, they show that incorporating news-derived signals improves forecast accuracy relative to models relying solely on load, weather and historical prices. The results underline that news captures regulatory, infrastructural and geopolitical information that is not directly observable in physical system data, reinforcing the complementarity between NLP-based features and established electricity market fundamentals (Bai et al., 2024, pp. 5–6).

Bi et al. (2025, pp. 1–2) further advance this line of research by introducing a large-scale, news-augmented benchmark dataset for electricity price forecasting using large language models. News texts are encoded using transformer-based embeddings and integrated into forecasting architectures across multiple time horizons (Bi et al., 2025, p. 6). Their empirical analysis demonstrates that NLP-augmented models deliver improvements in predictive performance, particularly during periods of market stress and heightened volatility (Bi et al., 2025, pp. 7 & 9). This study provides systematic evidence that textual representations derived from large language models are not merely auxiliary inputs but can become central components of modern electricity forecasting frameworks (Bi et al., 2025, pp. 2–3).

Despite the documented forecasting improvements, the reviewed literature reveals several recurring limitations that highlight the gap between statistical accuracy and market application. Early sentiment-based studies, such as Smales (2014) and Li et al. (2021), rely on aggregated indicators that compress heterogeneous information into a single numerical index, effectively obscuring the underlying qualitative drivers and restricting the model's ability to capture contextual nuances (Smales, 2014, pp. 277–278; Li et al., 2021, pp. 3–4). While more recent embedding-based frameworks improve semantic representation, they often introduce a "black-box" problem; Chakraborty et al. (2024, pp. 24–25) acknowledge that the high complexity of deep neural architectures like the REN framework reduces transparency, making it difficult to isolate which specific semantic components or events drive individual forecasts.

Furthermore, a significant gap exists regarding the economic translation of these signals. Although Guan et al. (2022), Bai et al. (2024), and Bi et al. (2025) demonstrate superior performance using statistical error metrics, they do not explicitly assess whether these gains translate into viable trading strategies under realistic execution constraints and transaction costs. This is particularly critical in short-term electricity markets, where the "informational decay" of news is rapid and market microstructures are unique (Weron, 2014, pp. 1031–1032). Finally, Bi et al. (2025, pp. 7–9) highlight concerns regarding stability and generalisation, noting that large language models can be sensitive to regime shifts and sample composition. This suggests that without a dynamic or unsupervised mechanism to account for evolving market vocabulary, models may fail to adapt to the rapidly changing power sector characteristics.

Taken together, the literature demonstrates a clear evolution from simple sentiment indicators towards embedding-based and event-driven forecasting frameworks. Textual information is increasingly treated as a quantitative input that complements historical prices and physical fundamentals. In energy markets in particular, where prices react strongly to expectations and policy-driven events, NLP-based forecasting models provide a natural extension to established time-series approaches discussed by Dannecker (2015) and Müsgens and Bade (2024). This body of research forms the conceptual foundation for the signal-generation framework developed in the subsequent Chapter 3.

### 2.3.2 From Forecasts to Trading Signals

While the literature reviewed in Section 2.3.1 focuses on improving predictive accuracy, forecasting performance alone does not automatically translate into economically meaningful trading outcomes. The transformation of forecasts into trading signals requires an additional decision layer that accounts for market frictions, risk constraints, and asymmetric payoff structures. In practice, market participants must determine when a forecast is sufficiently strong to justify a position and how it should be implemented within the specific microstructure of the traded market (Mack, 2014, pp. 7–9; Müsgens & Bade, 2024, pp. 22–23).

A common approach in the literature is to convert point forecasts or directional predictions into discrete trading actions, such as long, short, or neutral positions. This step typically involves the use of confidence thresholds or probability cut-offs, whereby positions

are only entered when the model's signal exceeds a predefined level of certainty. Such threshold-based decision rules help mitigate overtrading and reduce exposure to noise-driven forecasts, a consideration that is particularly important in volatile and illiquid segments of electricity markets (Z. T. Ke et al., 2019, pp. 8 & 18; Edwards, 2023, p. 270; Müsgens & Bade, 2024, pp. 33 & 44; Chakraborty et al., 2024, p. 7).

A closely related application of this forecast-to-signal paradigm can be found in electricity markets. Chen et al. (2025) analyse intraday electricity price forecasting from an explicitly economic perspective by embedding probabilistic price forecasts into concrete trading strategies, rather than evaluating models solely on statistical accuracy. Specifically, the authors translate forecasted price distributions into execution rules such as optimal selling strategies under fixed-volume and time constraints, thereby explicitly modelling the link between forecast uncertainty and trading decisions (J. Chen et al., 2025, pp. 1–3). Their empirical results show that improvements in forecast quality do not automatically lead to higher trading profits, as realised performance depends critically on execution timing, liquidity conditions, and market constraints in the intraday setting (J. Chen et al., 2025, pp. 18–19). These findings highlight that, even in electricity markets with rich physical fundamentals, the transformation of forecasts into tradable signals constitutes a distinct optimisation problem that must be assessed using economic performance criteria rather than predictive accuracy alone (J. Chen et al., 2025, p. 21).

Extending this forecast-to-execution separation beyond electricity markets, Kirtac and Germano (2024, pp. 3–4) apply a similar two-stage framework to the stock market by first producing sentiment-based directional predictions using transformer embeddings and subsequently applying confidence filters to derive trading positions. Methodologically, the authors derive sentiment scores from large language models that are fine-tuned using future excess returns as supervision, thereby directly linking textual information to expected price direction. Trading strategies are then constructed as long–short portfolios based on percentile thresholds of these sentiment scores, with transaction costs explicitly incorporated to assess economic viability under realistic market conditions. Their empirical results show that profitability is highly sensitive to signal calibration. Strategies with stricter confidence thresholds exhibit lower trade frequency but achieve superior risk-adjusted performance, as measured by Sharpe ratios and drawdown metrics (Kirtac & Germano, 2024, pp. 7–8).

In contrast to stock markets, the conversion from forecasts to trades in energy markets must also reflect physical and institutional constraints. Electricity prices are subject to sharp intraday reversals, negative pricing, and uneven liquidity across delivery periods (Section 2.1). As noted by Müsgens and Bade (2024, pp. 89 & 153) and Chakraborty et al. (2024, pp. 1 & 24), trading strategies that ignore these features risk systematic losses despite statistically accurate forecasts. Consequently, practical implementations often restrict trading to specific time windows, delivery products, or volatility regimes, aligning model outputs with operational feasibility. Interpretability plays a particularly important role in this context. Electricity traders frequently rely on discretionary judgement when reacting to unexpected events such as outages, regulatory announcements, or geopolitical developments. Therefore, qualitative understanding of why a price signal arises is often as important as the signal itself. Topic-based representations derived from c-TF-IDF, as introduced in Section 2.2.2, support this requirement by providing transparent keyword descriptions for each identified topic. This allows traders to distinguish, for example, between signals driven by "supply outages" and those linked to "macroeconomic conditions" (Grootendorst, 2022, pp. 3–4). Such interpretability facilitates discretionary overlays and reduces the risk of mechanically executing opaque model outputs.

A critical practical limitation in electricity markets concerns signal timeliness. As noted in Section 2.2.3, the computational costs associated with transformer-based models can introduce latency between news arrival and signal generation. In the intraday market, where price adjustments may occur within minutes of an information release, this creates a trade-off between semantic richness and signal velocity (Müsgens & Bade, 2024, p. 153; Edwards, 2023, p. 51). More complex models risk producing "stale" signals that decay before execution becomes economically viable, necessitating a balance between representational depth and computational efficiency in short-horizon contexts (Das, 2014, pp. 106–107; Jurafsky & Martin, 2025, p. 172).

Taken together, the literature demonstrates that the economic value of NLP-based forecasts depends critically on the design of the signal-generation layer rather than on predictive accuracy alone. Across both electricity and stock markets, effective strategies do not translate forecasts into trades mechanically but embed them within disciplined execution frameworks that account for confidence calibration, market microstructure, interpretability, and latency constraints. Empirical evidence shows that forecast improvements only become economically meaningful when combined with selective execution rules, risk

controls, and market-specific feasibility considerations. This distinction between statistical forecasting performance and tradable signal construction motivates the two-stage modelling approach adopted in this study. Consistent with the reviewed literature, NLP-derived forecasts are first generated and subsequently transformed into actionable long, short, or hold decisions under explicit decision rules, as detailed in Chapter 3.

# 3 Methodology

Chapter 3 outlines the empirical methodology employed to investigate whether energy-related news can be transformed into statistically and economically meaningful trading signals for the German electricity market. Building on the theoretical foundations presented in Chapter 2, this chapter details the end-to-end modelling pipeline, covering data collection, preprocessing, feature construction, model training, evaluation, and backtesting.

## 3.1 Framework Structure

This section introduces the methodological framework employed to transform energy-related news into actionable trading signals for the German electricity market. The overall design draws upon established principles in text-based financial forecasting, wherein unstructured textual data are transformed into structured numerical representations suitable for integration with traditional quantitative inputs (Das, 2014, pp. 3–5; Loughran & McDonald, 2020, pp. 1–3). In line with recent advances in electricity price forecasting (Bai et al., 2024; Bi et al., 2025), the framework combines transformer-based language models with gradient-boosted ensemble methods to capture both semantic content and non-linear interactions between news-derived and market-based features.

A distinctive feature of the proposed framework is its two-stage stacked ensemble architecture, which separates the tasks of spread direction forecasting and trading signal generation. This design reflects the distinction between statistical forecasting and tradable signal construction emphasised in the literature (Kirtac & Germano, 2024, pp. 3–4). The first stage employs XGBoost (T. Chen & Guestrin, 2016), a regularised gradient-boosting framework that has demonstrated strong performance in electricity price forecasting applications (Lago et al., 2018, pp. 395–397). XGBoost operates on the full feature set comprising baseline market variables and news-derived indicators, predicting a three-class target variable based on the difference between next-day spot and day-ahead auction prices. To ensure well-calibrated probabilities for downstream decision-making, Platt scaling is applied to the XGBoost outputs (Platt, 1999, pp. 63–66). The second stage employs LightGBM (G. Ke et al., 2017, pp. 1–2) as a meta-learner operating on a reduced feature space comprising baseline price features and calibrated XGBoost probability outputs. Following the stacking generalisation framework formalised by Wolpert (1992, pp.

241–242), the meta-learner learns optimal weightings of base model predictions, refining directional forecasts into actionable long/short/hold decisions. Prior to gradient-boosting optimisation, Ridge regression serves as a computationally efficient screening mechanism to identify promising time-decay parameter configurations (Hoerl & Kennard, n.d., pp. 55–58).

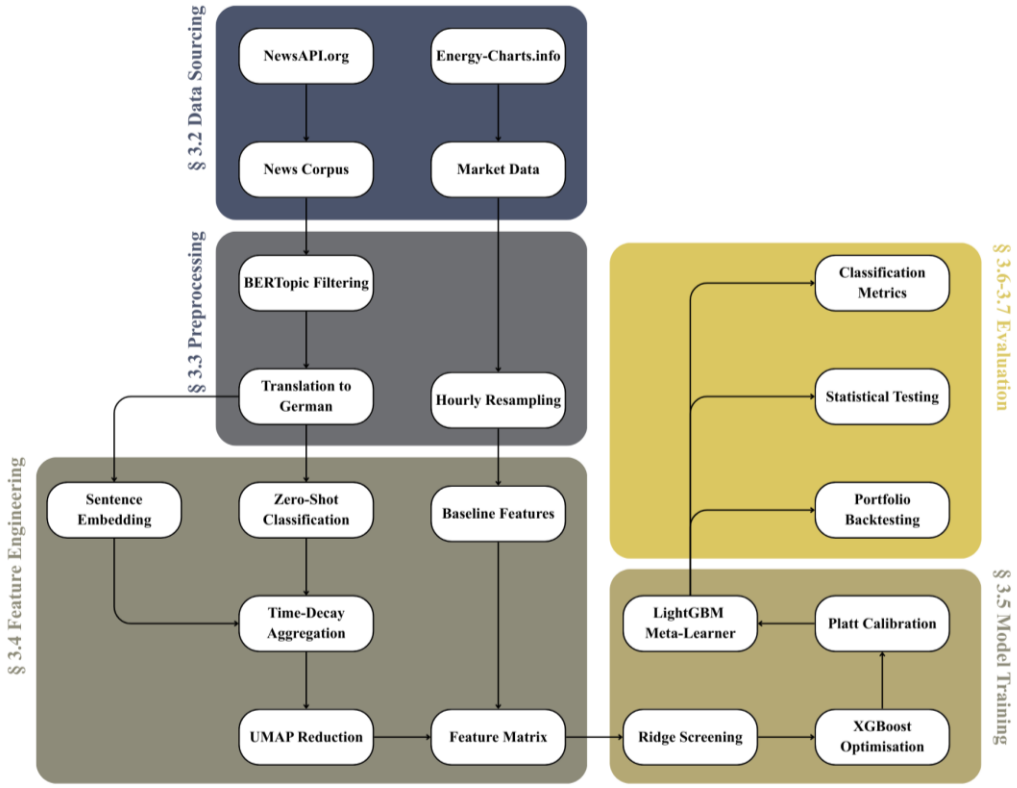


*Figure 1: Methodology Flowchart*

Figure 1 presents a condensed schematic of the full methodological pipeline, illustrating the flow from raw data acquisition through to economic performance evaluation. The pipeline comprises six principal stages: (i) data sourcing, which retrieves news articles and electricity market data; (ii) data preprocessing, which filters, translates, and aligns the heterogeneous data streams; (iii) feature engineering, which transforms raw inputs into baseline market features and NLP-derived indicators; (iv) model training, which optimises the two-stage ensemble architecture; (v) model evaluation, which assesses predictive performance using classification metrics and statistical tests; and (vi) backtesting, which evaluates economic viability under realistic trading assumptions.

The feature engineering stage produces two complementary categories of predictive inputs. Baseline features comprise market-derived variables capturing the structural characteristics of electricity price formation, including prices, generation and load volumes, lagged values at daily and weekly horizons, and cyclical temporal indicators (Weron, 2014, pp. 1049–1051). News-derived features are constructed via a dual NLP pipeline that extracts both discrete topical signals and continuous semantic representations. Topic classification employs zero-shot learning using a multilingual DeBERTa model (Laurer et al., 2024), whilst sentence embeddings are generated using a multilingual MiniLM model (Reimers & Gurevych, 2019; Wang et al., 2020). Both feature types undergo exponential time-decay aggregation to reflect the diminishing relevance of older news (Tetlock, 2007, pp. 1149–1151), followed by UMAP-based dimensionality reduction (McInnes et al., 2020).

The computational pipeline was implemented in Python and executed on a GPU-accelerated computing cluster. Parallel processing was employed throughout the feature engineering and model training stages, with particular attention to the parallel structure of the parameter grid search and cross-validation procedures. Generative AI tools were also used to support parts of the writing process and to assist with debugging during development. A detailed account of AI usage is provided in Appendix 10

## 3.2 Data Sourcing

This research project combines two synchronised data streams: (i) news headlines capturing energy-related information flows and (ii) German electricity market data capturing realised price and volume dynamics. The sourcing design prioritises coverage, timestamp quality, and hourly alignment, enabling subsequent aggregation of news signals to the market sampling frequency. Sections 3.2.1 and 3.2.2 describe the collection of news and market data, respectively.

### 3.2.1 News Data Collection

News data were collected using NewsAPI.org, an aggregation service providing programmatic access to historical and current news articles from diverse sources. NewsAPI.org was selected over alternative services due to its comprehensive metadata structure,

including timestamps, source attribution, and article descriptions, which enables precise temporal alignment with market data. Usage of such structured data consisting of online news articles and news streams has been employed in recent commodity market forecasting applications to extract market-relevant signals from high-frequency news flows (Chakraborty et al., 2024, pp. 2–3).

The news collection strategy encompassed both English- and German-language sources to capture the full spectrum of discourse surrounding German electricity markets. German-language sources provide domestic market insights, regulatory announcements, and regional policy developments, whilst English-language international sources offer broader market analysis, global energy trends, and institutional investor perspectives that influence local prices. This bilingual approach is particularly relevant given Germany's central role in European electricity markets, where cross-border flows and interconnector capacity create price dependencies across multiple national markets (Frauendorfer et al., 2018, pp. 2–3). All collected articles were subsequently translated into German (as detailed in Section 3.3.1) to enable consistent downstream natural language processing.

The collection period spanned from 1 November 2020 to 31 October 2025, providing a five-year window that captures significant market regime changes: the COVID-19 pandemic's demand disruptions, the 2022–2023 energy crisis triggered by geopolitical tensions following Russia's invasion of Ukraine, and the subsequent market stabilisation accompanied by accelerated renewable deployment (Graf et al., 2021, pp. 1, 6). This temporal coverage exposes the model to diverse market conditions, including both low-volatility and crisis periods, thereby improving robustness to the regime changes that characterise modern electricity markets.

The retrieval strategy employed daily queries via NewsAPI.org's premium subscription tier, which permits up to 100 articles per API call. Each day's query retrieved articles matching the specified keywords and source parameters (Table 1). On days when more than 100 matching articles were published, the API returned the 100 most relevant results ranked by NewsAPI's relevance algorithm; however, such instances were rare over the five-year period. This systematic daily collection methodology aligns with best practices for constructing news-based trading signals, where consistent temporal coverage prevents information gaps that could compromise signal quality (Smales, 2014, pp. 277–279). The

collection contained 89’517 English-language and 122’208 German-language articles before any preprocessing.

*Table 1: News Search Parameters*

| Parameter | English Sources | German Sources |
|---|---|---|
| News Outlets | ABC News, Associated Press, BBC News, Bloomberg, CNN, MSNBC, NBC News, Reuters, The Wall Street Journal, The Washington Post, Time, USA Today | Bild, Der Tagesspiegel, Die Zeit, Focus, Handelsblatt, Spiegel Online, WirtschaftsWoche |
| Keywords | ACER, Carbon, Climate, Coal, ECB, Electricity, Emissions, Energy, EPEX, European Comission, European Union, Gas, Grid, Hydro, IEA, Infrastructure, Interest Rates, Market, Nuclear, OECb, OPEC, Oil, Pipeline, Policy, Renewable, Solar, Storage, Supply, Tariffs, Transmission, Wind | EPEX, Erdgas, Erdöl, EU, EZB, IEA, Kohle, OPEC, Strombedarf, Stromeffizienz, Stromerzeugung, Stromnachfrage, Stromsparen, Wetter, Zentralbank |
| Articles Retrieved | 89’517 | 122’208 |
| Timeframe | 1 November 2020 - 31 October 2025 | 1 November 2020 - 31 October 2025 |

The keyword selection strategy balanced domain specificity with comprehensive coverage by incorporating terminology related to energy infrastructure, market mechanisms, relevant institutional abbreviations, and macroeconomic indicators. This multi-faceted approach recognises that electricity prices are shaped by heterogeneous information, ranging from system fundamentals such as renewable generation (and forecast) conditions to broader macroeconomic signals that proxy changes in industrial activity and electricity demand (Edwards, 2023, pp. 51 & 114).

### 3.2.2 Energy Market Data Collection

Energy market data were sourced from Energy-Charts.info, a platform maintained by the Fraunhofer Institute for Solar Energy Systems (ISE). The platform aggregates data from multiple authoritative sources, including ENTSO, EPEX, and EEX Transparency Platform, providing harmonised datasets that would otherwise require complex reconciliation across different source systems (Fraunhofer ISE, 2025).

Data were retrieved at 15-minute granularity for the period 1 November 2020 to 31 October 2025, matching the news collection timeframe. The following price variables were collected: day-ahead auction prices (EUR/MWh) representing the uniform clearing prices established in the daily EPEX auction, and intraday continuous market prices

(EUR/MWh) reflecting real-time trading activity. Energy-Charts.info already matches the day-ahead auction price to its corresponding spot price time slot. These two price series form the basis for computing the spot-day-ahead spread that serves as the target variable for this research project, as described in Section 3.4.1.

Power generation volumes were retrieved, disaggregated by source category to capture the supply-side dynamics that influence price formation. The dataset includes output from onshore wind, offshore wind, solar photovoltaic, nuclear, lignite, hard coal, natural gas, pumped-hydro, and run-of-river hydro facilities. This granular breakdown enables the model to account for the merit-order effects discussed in Section 2.1, where low-marginal-cost renewables displace higher-cost thermal generation and compress electricity prices during periods of high renewable feed-in. Total electricity demand (MW) was also collected, representing the aggregate load that must be balanced against generation in real time.

The raw dataset comprised approximately 175’000 observations at 15-minute resolution. These data were converted from the original JSON format to CSV for subsequent preprocessing. Quality assessment of the raw data identified several categories of missing or anomalous values. Gaps in price reporting occurred during approximately 0.4% of 15-minute intervals.

## 3.3 Data Preprocessing

The preprocessing pipeline comprised two parallel workflows: one for news data and one for energy market data, which were subsequently integrated to construct the final analytical dataset. This section details the procedures employed to ensure data quality, topical relevance, and temporal alignment across these heterogeneous sources.

### 3.3.1 News Corpus Refinement

The broad keyword-based retrieval strategy employed in Section 3.2.1, whilst ensuring comprehensive coverage of potentially relevant content, inevitably captured substantial quantities of off-topic material. Articles matching terms such as "energy" or "market" frequently addressed topics unrelated to electricity trading, including sports events,

entertainment news, and general political coverage. Rigorous filtering was therefore required to isolate energy-relevant news content suitable for signal generation.

Off-topic content was identified and removed through an iterative refinement procedure using BERTopic (Grootendorst, 2022, pp. 1–3), an unsupervised topic modelling framework described in Section 2.2.2 BERTopic was selected for this task because it enables the discovery of latent thematic structures without requiring predefined category labels, making it well-suited to exploratory corpus analysis in which the full range of off-topic content is not known in advance (Grootendorst, 2022, pp. 7–8). The framework's use of HDBSCAN clustering also accommodates topics of varying sizes and explicitly identifies noise documents that do not belong to any coherent theme.

Each iteration of the refinement procedure proceeded as follows: (i) a topic model was fitted to the corpus to discover latent thematic clusters; (ii) human reviewers inspected topic labels, representative keywords, and sample documents to identify non-relevant themes; (iii) characteristic keywords and phrases associated with off-topic clusters were extracted to construct exclusion filters; (iv) articles containing these keywords in titles or descriptions were removed from the corpus; and (v) the filtered corpus was resubmitted for subsequent refinement. This human-in-the-loop approach ensured that automated clustering was validated against domain expertise before content removal, preventing the erroneous exclusion of legitimately relevant articles that might share surface-level vocabulary with off-topic content.

Nine iterative filtering rounds were applied to the English corpus, progressively removing articles associated with sports coverage (team names, match results, athlete profiles), entertainment news (film releases, celebrity coverage, television programming), crime reporting (arrests, trials, investigations), retail and consumer news (product launches, shopping trends), health and medical coverage unrelated to energy policy, weather reporting without market implications, and general political coverage lacking energy or economic relevance. This process reduced the English corpus from 89’517 to 39’744 articles, a 55.6% reduction. The substantial reduction reflects the broad semantic scope of English-language international news sources, where energy-related keywords frequently co-occur with unrelated content.

The German corpus required only three iterative rounds, targeting crime, education, sports, and tourism content, and was reduced from 122’208 to 113’894 articles, which

represents a more modest 6.8% reduction. The disparity in reduction rates suggests that German-language sources retrieved via energy-specific keywords were more topical than their English-language counterparts. This likely reflects the more specialised focus of German business publications such as Handelsblatt and WirtschaftsWoche, which concentrate on economic and market coverage rather than general news. A detailed log of excluded topic categories and representative keywords per iteration is provided in Appendix 1.

Following topic-based filtering, English articles were translated into German using the Google Translate API via the deep-translator Python library. Translation into German (rather than translating German content into English) was chosen for two reasons. First, German-language transformer models fine-tuned on energy and economic terminology prevalent in the DACH-region discourse were available for downstream classification tasks. Second, the majority of the filtered corpus (74%) was already in German, minimising the proportion of content subject to potential translation errors. The translated English corpus (39’744 articles) was merged with the filtered German corpus (113’894 articles), yielding a total of 153’638 articles. After removing translation failures and malformed entries, the final corpus comprised 141’716 German-language articles spanning the five-year collection period.

### 3.3.2 Energy Market Data Processing

Energy market data underwent preprocessing to ensure temporal alignment with the news corpus and to prepare variables for feature engineering. The raw data, retrieved at 15-minute granularity, were resampled to hourly frequency by computing means for price variables and sums for volume variables. The hourly spot price aggregation was selected to align with the day-ahead auction structure, where prices are determined for each hour of the following day, and to reduce computational overhead whilst preserving meaningful intraday variation. Following resampling, the energy dataset comprised 43’990 hourly observations spanning the five-year collection period.

Missing values were identified in 192 hourly records, primarily during daylight saving time transitions and periods of data reporting gaps. These observations were excluded rather than imputed to preserve data integrity, as interpolation of price data during market discontinuities could introduce spurious patterns. Anomalous price observations,

including negative prices below -500 EUR/MWh and favourable prices exceeding 1’000 EUR/MWh, were retained in the dataset, as these extreme values represent genuine market conditions during periods of renewable oversupply or acute scarcity rather than data errors. The final processed energy dataset comprised 43’798 hourly observations.

## 3.4 Feature Engineering

The feature engineering process transforms raw energy market data and pre-processed news articles into a structured feature matrix suitable for forecasting modelling. This section details the construction of baseline features, the extraction of topical signals via zero-shot classification, the generation of semantic embeddings, and their temporal aggregation via exponential decay weighting.

### 3.4.1 Baseline Features

The baseline feature set comprises variables derived directly from the energy market data described in Section 3.2.2. These features capture the fundamental supply-demand dynamics that drive electricity price formation, as well as temporal patterns that reflect the cyclical nature of electricity consumption.

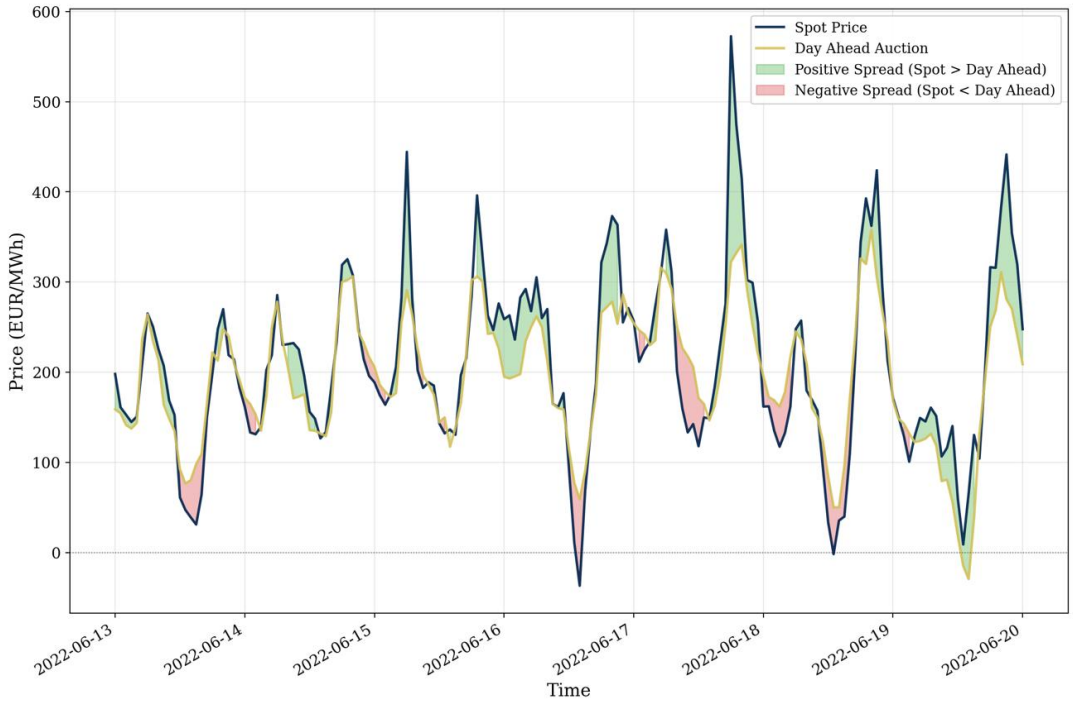


*Figure 2: Spot-, Day Ahead Auction Price of a sample week*

**Price and Spread Features**: The primary price variables are the day-ahead auction price and the intraday spot price, resampled at hourly frequency. The absolute spread between these prices serves as the target variable for classification. Observations are labelled as Long (positive spread, spot > day-ahead), Short (negative spread, spot < day-ahead), or Neutral when the absolute spread falls within a symmetric deadband of ±3.0 EUR/MWh, which is based on the observed average Bid-Ask spread in the markets (Müsgens & Bade, 2024, p. 20). To aid interpretation of the spread definition, Figure 2 provides a representative example of both series and their resulting spread.

**Generation and Load Features**: Aggregate electricity generation and total system load (MW) provide direct measures of supply–demand balance. As discussed in Section 2.1.4, the merit-order effect implies that price formation depends critically on the marginal generating unit dispatched to meet demand (Edwards, 2023, pp. 134–138).

**Lagged Features**: To capture autocorrelation in electricity prices and demand patterns, lagged values are included at two horizons: 24 hours and 168 hours. Specifically, the feature set contains *price_lag_24*, *price_lag_168*, *load_lag_24*, and *load_lag_168*. These lags reflect the strong daily and weekly seasonality in electricity consumption patterns, as discussed in Section 2.1.4, in which demand exhibits pronounced differences between weekdays and weekends and between daytime peak hours and overnight off-peak periods (Weron, 2014, pp. 1049–1051).

**Temporal Features**: Categorical and cyclical time indicators encode the position of each observation within relevant periodic structures:

- *hour* (0–23): Captures intraday variation aligned with consumption patterns
- *day_of_week* (0–6): Distinguishes weekday from weekend dynamics
- *day_of_year* (1–366): Encodes annual seasonal effects, including heating and cooling demand
- *week_of_year* (1–52): Provides coarser seasonal resolution
- *month* (1–12): Captures broader seasonal trends in renewable generation and demand

Together, these 13 baseline features establish a foundation that captures the structural characteristics of electricity markets before the incorporation of news-derived signals.

### 3.4.2 Topic Classification via Zero-Shot Learning

News articles are transformed into structured topical features via a zero-shot classification pipeline that assigns each article to predefined energy-relevant categories derived from Section 2.1. This approach leverages the semantic generalisation capabilities of NLP models, as discussed in Section 2.2.2, enabling classification without task-specific training data (Yin et al., 2019, pp. 1–3).

The classification employs the mDeBERTa-v3-base-xnli-multilingual-nli-2mil7 model (Laurer et al., 2024), which extends Microsoft's DeBERTa-v3 architecture (He et al., 2023) to multilingual zero-shot classification. The model outputs probabilities for each label, and the label with the highest probability is assigned as the classification. This enables flexible classification across arbitrary label sets without retraining (Laurer et al., 2024, pp. 3–6). The classification uses the following hypothesis template:

*"Der Artikel handelt von: {label}."*

*Table 2: Zero-Shot Topic Taxonomy*

| Category | Leaf Topics |
|---|---|
| **Nachfrage (Stromverbrauch)** | *Steigender Stromverbrauch durch Wirtschaft, Industrie oder Extremwetter; Sinkender Stromverbrauch durch Konjunkturschwäche oder mildes Wetter* |
| **Angebot (Erzeugung & Infrastruktur)** | *Kraftwerksausfälle, Netzengpässe oder geringe erneuerbare Einspeisung; Hohe erneuerbare Einspeisung, neue Kapazitäten oder stabile Netze* |
| **Brennstoffpreise** | *Steigende Gas-, Kohle- oder $CO_2$-Preise; Fallende Gas-, Kohle- oder $CO_2$-Preise* |
| **Wetter** | *Kaltes, windarmes oder bewölktes Wetter; Mildes Wetter, starke Winde oder viel Sonneneinstrahlung* |
| **Wirtschaft & Konjunktur** | *Positive Wirtschaftsentwicklung, steigende Industrieproduktion oder Unternehmenswachstum; Rezession, Unternehmenskrise oder sinkende Industrieproduktion* |
| **Finanzmärkte & Geldpolitik** | *Zinsentscheidungen, Inflation oder Währungsschwankungen; Börsennachrichten, Unternehmensgewinne oder Investitionen* |
| **Handel & Aussenwirtschaft** | *Zölle, Handelskonflikte oder Exportbeschränkungen; Handelsabkommen, Marktöffnung oder Lieferkettenentwicklung* |
| **Geopolitik & Konflikte** | *Krieg, Sanktionen, Terrorgefahr oder geopolitische Spannungen; Friedensgespräche, Diplomatie oder Aufhebung von Sanktionen* |
| **Technologie & Industrie** | *Technologieentwicklung, Halbleiter, Elektromobilität oder Batterieproduktion; Industriepolitik, Produktionsstandorte oder Unternehmensstrategien* |
| **Politik & Regulierung** | *Energiepolitik, Klimagesetze oder EU-Regulierung; Innenpolitik, Regierungsbildung oder Wahlen* |
| **Sonstiges** | *Sport, Unterhaltung, Kultur oder Lokalnachrichten ohne Wirtschaftsbezug* |

The topic structure comprises eleven high-level categories reflecting distinct channels through which news may influence electricity prices. Each category (except "Sonstiges") contains two directionally opposed leaf topics: one representing conditions that would

exert upward pressure on prices and one representing conditions that would ease prices, enabling the model to capture both topical relevance and the implied price-impact direction. The complete taxonomy is presented in Table 2.

Each article's title is first classified against the eleven category labels. If the highest-scoring category exceeds the Stage 1 threshold ($\tau_1$ = 0.25), the article proceeds to Stage 2 classification within that category; otherwise, it is routed to "Sonstiges." In Stage 2, a threshold of $\tau_2$ = 0.20 governs whether the specific leaf topic is retained. Articles initially classified as "Sonstiges" undergo secondary classification using the article description, providing a fallback for ambiguous headlines.

The threshold values were determined through manual inspection of classification score distributions, balancing coverage against precision (Laurer et al., 2024, pp. 5–7). Articles ultimately classified as having no energy relevance ("Sonstiges") are excluded from subsequent aggregation.

### 3.4.3 Sentence Embedding Generation

Complementing discrete topic classification, dense sentence embeddings capture semantic information that predefined categories may not fully represent. As introduced in Section 2.2.2, sentence embeddings encode the full semantic content of text as continuous vectors, preserving nuanced information that discrete labels cannot capture.

The embedding pipeline employs the paraphrase-multilingual-MiniLM-L12-v2 model (Reimers & Gurevych, 2019, pp. 1–2), which builds on the MiniLM knowledge distillation framework (Wang et al., 2020, pp. 2–5) to compress larger transformer models while preserving most of their representational capacity. This multilingual MiniLM model generates 384-dimensional vector representations optimised for semantic similarity across 50+ languages, including German. For each article, the embedding is computed from the title using mean pooling over contextualised token representations, prioritising computational efficiency while capturing primary informational content.

### 3.4.4 Time-Decay Aggregation of News Features

Individual article-level features must be aggregated to the hourly resolution of the energy market data. The aggregation method in this research project employs exponential time-decay weighting, assigning greater influence to recent articles while retaining diminishing contributions from older content. This reflects the intuition that news impact decays as information is absorbed into prices (Tetlock, 2007, pp. 1149–1151), which effect has also been observed in the commodity markets (Li et al., 2021, pp. 3–4).

In electricity markets, the high frequency of price dynamics and the rapid incorporation of information suggest that news relevance diminishes relatively quickly. Forecast revisions for renewable generation, updates to plant availability, and shifts in demand expectations are continuously reflected in intraday trading, implying that older news articles provide progressively less marginal information (Weron, 2014, pp. 1031–1032).

The exponential decay formulation assigns weight $w_i = e^{-\lambda \cdot h_i}$ to article $i$, where $\lambda$ is the decay rate parameter and $h_i$ is the number of hours since publication. For each hourly timestamp $t$, time-decayed weighted counts are computed for each of the leaf topics by summing decay-weighted indicators across all articles published within a lookback window $L$. Articles classified as "Sonstiges" are excluded. Embedding aggregation follows the same procedure, using weighted averaging rather than summation to produce a single 384-dimensional vector per timestamp.

The lookback window and decay rate jointly determine the temporal dynamics of aggregation. A grid search evaluates 30 parameter combinations:

*Table 3: Parameter Grid for Time-Decay Aggregation*

| Parameter | Values tested |
|---|---|
| *Lookback Window (L)* | 24, 48, 72, 168, 336, 504 hours |
| *Decay Lambda (λ)* | 0.01, 0.05, 0.1, 0.25, 0.5 |

The optimal combination is selected through the model validation procedure described in Section 3.5. The default parameters ($L$ = 336 hours, $\lambda$ = 0.05) reflect a two-week lookback with moderate decay; an article published 24 hours ago retains approximately 30% of its initial weight, while an article from one week ago retains less than 1%.

### 3.4.5 Dimensionality Reduction of Aggregated Embeddings

The 384-dimensional time-decayed embedding vectors pose challenges for downstream modelling due to high dimensionality and overfitting risk. UMAP, introduced in Section 2.2.2, is employed for dimensionality reduction, offering superior scalability and global structure preservation compared to alternatives (McInnes et al., 2020, pp. 1–2).

The UMAP configuration employs the following parameters:

- ***n_components* = 20**: Target dimensionality, reducing the 384-dimensional input by a factor of approximately 19
- ***n_neighbors* = 15**: Local neighbourhood size, controlling the balance between local and global structure preservation
- ***min_dist* = 0.1**: Minimum distance between points in the embedding, governing clustering tightness

These parameters follow the guidelines provided by McInnes et al. (2018), who recommend *n_neighbors* values between 10 and 50 to balance local and global structure preservation, and *min_dist* values between 0.0 and 0.5 to control the tightness of the resulting clusters. The choice of 20 components provides sufficient expressiveness for downstream prediction while substantially reducing the dimensionality of the feature space.

The final output is a matrix of dimension $(T \times 20)$, where T denotes the number of hourly observations in the analysed period and each row represents the reduced embedding for a single hourly timestamp. These 20 embedding dimensions, combined with the 14 topic count features, yield 34 news-derived features per parameter combination.

As noted in Section 2.2.3, the stochastic nature of UMAP implies that strict determinism would require a fixed random seed. Given the size of the aggregated embedding dataset, dimensionality reduction was executed in parallel to ensure computational feasibility, which precluded the use of a single global seed without enforcing a single-threaded pipeline. This introduces run-to-run variation in the reduced representations.

### 3.4.6 Feature Integration & Scaling

The complete feature matrix integrates baseline features with news-derived features to produce the input representation for predictive modelling. Table 4 summarises the complete feature set:

*Table 4: Summary of Feature Set*

| Feature Category | Count | Description |
|---|---|---|
| *Price Features* | 2 | Day-ahead price, spot price |
| *Spread Features* | 1 | Absolute spread |
| *Load Features* | 1 | System load |
| *Lagged Features* | 4 | 24h and 168h lags for price and load |
| *Temporal Features* | 5 | Hour, day of week, day/week/month of year |
| *Topic Features* | 14 | Time-decayed counts per leaf topic |
| *Embeddings Features* | 20 | UMAP-reduced embedding dimensions |

The news-derived features (34 dimensions) account for approximately 72% of the total feature space, reflecting the research focus on extracting predictive signals from textual information.

The dataset is partitioned chronologically to preserve the temporal ordering essential for time-series forecasting (Hyndman & Athanasopoulos, 2021, pp. 192–195):

- **Training Set**: First 70% of observations
- **Validation Set**: Next 20% of observations
- **Test Set**: Final 10% of observations

This temporal separation ensures that all test-set observations occur strictly after all training observations, reflecting the realistic constraint that forecasting models cannot access future information. The validation set serves for hyperparameter tuning and time-decay parameter selection, while the test set provides an unbiased estimate of out-of-sample performance.

## 3.5 Model Training and Optimisation

This section details the hierarchical training methodology employed to transform engineered features into actionable trading signals. The framework implements a two-stage stacked ensemble architecture wherein an XGBoost classifier first learns complex interactions between news-derived features and market dynamics, followed by a LightGBM meta-learner that refines these predictions into long/short/hold trading decisions. This

approach draws on established stacking generalisation principles, in which base-level predictions serve as meta-features for subsequent learners, thereby improving overall predictive accuracy (Wolpert, 1992, pp. 241–259).

### 3.5.1 Training Framework Architecture

The training pipeline follows a systematic progression through five computational stages, designed to maintain strict temporal ordering whilst maximising parallel efficiency where dependencies permit. Building upon the pre-processed features described in Section 3.4, the workflow proceeds as follows.

The two-stage stacked architecture addresses a fundamental challenge in news-driven electricity forecasting: the need to extract predictive signal from high-dimensional, noisy textual features whilst avoiding overfitting on limited training samples. As discussed in Section 2.2.3, limitations of NLP in financial applications include the risk of spurious correlations when combining textual and numerical features. The hierarchical approach mitigates this by progressively reducing dimensionality across stages.

The first stage employs XGBoost as the base learner because it is effective in electricity price forecasting, where ensemble methods consistently outperform single-model alternatives (Lago et al., 2018, pp. 395–297). XGBoost's regularised gradient-boosting framework provides implicit feature selection that identifies which news-derived signals contain genuine predictive content. However, careful regularisation is required to prevent spurious correlations when combining BERT-based news features with price series (Guan et al., 2022, pp. 3573–3575).

The second stage employs LightGBM as the meta-learner, operating on a substantially reduced feature space comprising baseline price features and calibrated XGBoost probability outputs. This dimensionality reduction aims to mitigate the risk of overfitting whilst enabling the meta-learner to learn optimal weightings of the base model's predictions across different market conditions. The approach aligns with the stacking generalisation framework formalised by Wolpert (1992), in which base-level predictions serve as meta-features for subsequent learners, thereby improving overall predictive accuracy.

The full workflow proceeds through the following stages:

1. **Feature Parameter Selection**: Evaluate 30 time-decay parameter combinations using Ridge regression with cross-validation (CV), selecting the top five configurations for subsequent model training.
2. **XGBoost Hyperparameter Optimisation**: Conduct randomised search across 80 hyperparameter configurations for each selected parameter combination, employing expanding-window CV to preserve temporal dependencies.
3. **Probability Calibration**: Apply Platt scaling to transform XGBoost output scores into well-calibrated probability estimates suitable for meta-feature generation.
4. **Meta-Learner Training**: Train LightGBM classifiers on both signal-enriched features (baseline + XGBoost predictions) and baseline-only features for comparative analysis.

The chronological data split established in Section 3.4.6 is maintained throughout, ensuring that all model development decisions are made without access to future observations.

### 3.5.2 Feature Parameter Selection via Ridge Regression

Before committing substantial computational resources to gradient boosting optimisation, the pipeline implements a lightweight screening phase using Ridge regression classifiers. This approach efficiently identifies promising time-decay parameter combinations from the 30-element grid (6 lookback windows × 5 decay rates) described in Section 3.4.4.

Ridge regression incorporates L2 regularisation to constrain coefficient magnitudes, mitigating multicollinearity in high-dimensional feature spaces (Hoerl & Kennard, n.d., pp. 55–58). The regularisation strength $\alpha$ is selected via grid search over 13 logarithmically spaced values spanning $[10^{-3}, 10^{3}]$. For each of the 30 parameter combinations, Ridge classifiers are fitted using 5-fold CV with accuracy as the scoring metric.

Following cross-validation, the five parameter combinations achieving the highest validation accuracy proceed to XGBoost training. This two-stage screening approach serves dual purposes: (i) reducing computational burden by eliminating poorly performing configurations early, and (ii) enabling comparative analysis of how different temporal aggregation strategies influence predictive performance. The efficiency of linear models for

initial parameter screening has been demonstrated in similar ensemble pipelines (Caruana et al., 2004, pp. 3–5).

### 3.5.3 XGBoost Hyperparameter Optimisation

XGBoost implements regularised gradient boosting with built-in L1 and L2 penalties that control model complexity (T. Chen & Guestrin, 2016, pp. 785–789). The pipeline employs randomised search sampling 80 configurations from the distributions specified in Table 5, following Bergstra et al. (2012, pp. 288–290), who demonstrate that random search efficiently explores high-dimensional parameter spaces.

*Table 5: Parameter Search Settings for XGBoost*

| Parameter | Distribution | Range | Purpose |
|---|---|---|---|
| *n_estimators* | Uniform integer | [100, 400] | Number of boosting rounds |
| *max_depth* | Uniform integer | [2, 6] | Maximum tree depth |
| *learning_rate* | Log-uniform | [0.01, 0.15] | Step size shrinkage |
| *subsample* | Uniform | [0.5, 0.9] | Row sampling ratio |
| *colsample_bytree* | Uniform | [0.5, 0.9] | Column sampling ratio |
| *gamma* | Uniform | [0.5, 5.0] | Minimum loss reduction for split |
| *min_child_weight* | Uniform integer | [3, 12] | Minimum instance weight in child |
| *reg_alpha* | Log-uniform | [$10^{-3}$, $10^{1}$] | L1 regularisation term |
| *reg_lambda* | Log-uniform | [$10^{-2}$, $10^{2}$] | L2 regularisation term |

The distributions are deliberately constrained relative to typical defaults to address overfitting in time-series contexts with limited training samples. Temporal ordering is preserved through an expanding-window (walk-forward) validation scheme with eight folds, a 72-hour step size, and a 336-hour minimum training duration (Tashman, 2000, pp. 440–445).

Class imbalance is addressed using squared inverse-frequency sample weighting, amplifying the contribution of minority-class observations to the loss function (Haibo He & Garcia, 2009, pp. 1270–1275). Early stopping monitors validation-set multi-class log-loss and terminates training after 20 iterations without improvement. The macro-averaged F1 score serves as the primary optimisation target, ensuring equal weight across classes regardless of prevalence (Prechelt, 1998, pp. 761–763):

$$F_{1,\mathrm{macro}} = \frac{1}{K}\sum_{c=1}^{K} \frac{2 \cdot \mathrm{Precision}_c \cdot \mathrm{Recall}_c}{\mathrm{Precision}_c + \mathrm{Recall}_c}$$

This choice reflects the trading application's requirement for reliable signals across all market conditions. Macro-averaging ensures that minority-class performance receives equal weight in model selection, preventing the optimiser from sacrificing Neutral-class accuracy to improve performance on the more prevalent Long and Short classes (Sokolova & Lapalme, 2009, pp. 430–431).

### 3.5.4 Probability Calibration via Platt Scaling

Whilst XGBoost with the *softprob* objective produces normalised probability vectors, these outputs may not be well-calibrated in the sense that a predicted probability of 0.7 corresponds to 70% empirical accuracy (Niculescu-Mizil & Caruana, n.d., pp. 2–4). Poorly calibrated probabilities can degrade downstream model performance when used as meta-features, as the LightGBM meta-learner may learn to compensate for systematic biases rather than exploiting genuine predictive signal.

Platt scaling converts a classifier's raw scores into calibrated probabilities by fitting a sigmoid mapping on a held-out calibration set (Platt, 1999, pp. 63–66). This approach is well-suited to gradient-boosting ensembles, which often exhibit systematic over- or underconfidence (Niculescu-Mizil & Caruana, n.d., pp. 1–2).

The pipeline applies Platt scaling to the best-performing XGBoost model using the following configuration:

- **Method**: Sigmoid (Platt scaling, as opposed to isotonic regression)
- **Cross-validation mode**: *cv='prefit'*, to indicate the base model is already trained
- **Calibration set**: Held-out validation partition (20% of data)

Using the prefit mode applies calibration without retraining the base classifier, fitting the sigmoid parameters directly to validation-set predictions. This approach avoids the computational cost of refitting XGBoost whilst still correcting for probability miscalibration. The calibrated model replaces the uncalibrated version for all downstream operations, including meta-feature generation for LightGBM training.

### 3.5.5 Meta-Learner Training with LightGBM

LightGBM employs gradient-based sampling and feature bundling optimisations that achieve substantial speedups over traditional gradient-boosted decision tree implementations whilst maintaining predictive accuracy (G. Ke et al., 2017). These efficiency gains make LightGBM well-suited to the meta-learner role, where rapid iteration across parameter configurations is required.

The training framework implements stacked generalisation (Wolpert, 1992) by using XGBoost predictions as meta-features for LightGBM. In this two-level architecture, the base learner (XGBoost) generates predictions that serve as inputs to the meta-learner (LightGBM), thereby enabling the ensemble to learn complex combinations of base-model outputs. Specifically, four features are extracted from calibrated XGBoost output: the predicted class label (-1, 0, or 1) and the probability estimates for the Neutral, Long, and Short classes. These meta-features capture both the hard prediction and the confidence distribution, enabling the meta-learner to weight XGBoost contributions appropriately. Research on stacking ensembles demonstrates that probability outputs typically yield superior meta-features compared to class labels alone, as they encode uncertainty information that is lost in hard predictions (Džeroski & Ženko, 2004, pp. 266–268).

To quantify the contribution of news-derived signals, two LightGBM models are trained in parallel. The Signal Model combines baseline price features with XGBoost meta-features, representing the full predictive pipeline that incorporates news-derived information. The Baseline Model uses price features only, providing a reference for ablation analysis that isolates the marginal contribution of news signals. Comparing these models provides direct evidence for whether news-derived signals add predictive value beyond price history; if the Signal Model significantly outperforms the Baseline, this validates the news feature engineering methodology.

Unlike the XGBoost randomised search, LightGBM employs a grid search over a more constrained parameter space, reflecting the meta-learner's role in refinement rather than primary feature learning. Such a search is feasible given the meta-learner's reduced feature space (7 inputs versus 47 for XGBoost), where the smaller search volume permits complete enumeration without the sampling variance inherent in random search. Table 6 presents the parameter grid:

*Table 6: LightGBM Grid Search Parameters*

| Parameter | Values | Description |
|---|---|---|
| *num_leaves* | [31, 50, 100] | Maximum leaves per tree |
| *max_depth* | [5, 10, 15] | Maximum tree depth |
| *learning_rate* | [0.01, 0.05, 0.1] | Boosting step size |
| *n_estimators* | [100] | Fixed boosting rounds |
| *min_child_samples* | [20] | Minimum samples per leaf |
| *subsample* | [0.8] | Row sampling ratio |
| *colsample_bytree* | [0.8] | Column sampling ratio |

LightGBM training mirrors the XGBoost configuration described in Section 3.5.3, including expanding-window cross-validation with 5 folds (reduced from 8 to reflect the simpler optimisation landscape), a 72-hour step size, and a 336-hour minimum training size. Early stopping terminates training after 20 iterations without improvement. Class imbalance is addressed using inverse-frequency sample weights computed from the training set's class prevalence.

## 3.6 Model Evaluation Framework

The evaluation framework compares the Signal model (incorporating news-derived features) against the Baseline model (utilising price-based features only) through classification metrics, statistical significance testing, and probabilistic assessment. This section details the methodology underpinning the evaluation strategy.

### 3.6.1 Classification Metrics and Statistical Testing

Given the three-class prediction task (Long, Neutral, Short), performance is evaluated using the macro-averaged F1 score as defined in Section 3.5.3, which ensures equal weighting across classes regardless of prevalence. Per-class precision, recall, and F1-scores are reported to diagnose systematic misclassification, complemented by confusion matrices that visualise error structure.

For probabilistic discrimination, the Area Under the Receiver Operating Characteristic Curve (AUC) is computed via a One-versus-One (OvO) decomposition across the $\frac{K(K-1)}{2}$ class pairs, with macro-averaging to avoid favouring majority classes.

To test whether observed performance gaps between the Signal and Baseline models are attributable to random variation, McNemar's test is applied to paired predictions on identical instances (McNemar, 1947, pp. 153–155). Let *b* denote instances correctly classified only by the Signal model and *c* those correctly classified only by the Baseline model; the continuity-corrected statistic is:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

Under the null of equal error rates, the statistic follows a chi-square distribution with one degree of freedom. Focusing on discordant pairs increases power relative to aggregate accuracy comparisons, aligning with the evidence-based evaluation philosophy articulated in Section 4.3.

### 3.6.2 Temporal Validation and Class Imbalance Handling

The evaluation utilises the same expanding-window split described in Section 3.5.3 to ensure consistency between training and testing environments. The held-out test set remains the primary guardrail against leakage.

Metrics are computed using the squared inverse-frequency sample weights defined in Section 3.5, so that evaluation mirrors the training loss. Neutral-class probability thresholds are tuned on the validation folds (without retraining) to rebalance precision and recall; methodological details are provided in Section 3.5 to avoid duplication. Feature importance is reported via LightGBM gain rankings, with SHAP values as a complementary local explanation tool (Lundberg & Lee, 2017, pp. 1–2).

## 3.7 Backtesting Framework

This section specifies the backtesting framework employed to evaluate the economic implications of the model-generated trading signals developed in Section 3.5 and 3.6. The implementation is fully deterministic and operates exclusively on the held-out test sample established in Section 3.4.6, translating discrete model outputs into executable positions and subsequently into profit-and-loss (P&L) under explicit assumptions regarding timing, position sizing, and transaction costs. The backtest is designed as a signal-to-portfolio

pipeline, consistent with best practice in empirical asset pricing and systematic trading research, wherein the mapping from predictions to trades is fixed ex ante and performance is measured out of sample (Campbell et al., 2012, pp. 155–159).

### 3.7.1 Trading Strategy Specification

The LightGBM classifiers trained in Section 3.5.5 yield a discrete action $a_t \in -1,0,+1$ at each hourly decision point $t$, corresponding to the three-class target variable defined in Section 3.4.1: $+1$ denotes a Long signal, $-1$ a Short signal, and $0$ a Neutral (no-trade) decision. In addition to the two model-based strategies this research project also evaluates a naïve benchmark that takes a Long position at every decision point (i.e. $\mathrm{a_t} \equiv +1$ ) to provide a simple reference level of performance. These actions are translated into trades against the realised spread series which serves as the basis for P&L calculation. Where a horizon-aligned spread series is provided in the dataset, this series is used directly; otherwise, the spread is computed from contemporaneous prices. Periods with undefined spread values (e.g., due to shifting at the end of the sample) are treated as non-trading periods, and the portfolio is carried forward unchanged. The mapping from predictions to actions is fixed prior to evaluation and applied uniformly across candidate strategies to preserve out-of-sample integrity (Campbell et al., 2012, pp. 149–151; De Prado, 2018, pp. 190–193).

Economic performance is evaluated using a self-financing portfolio simulation that commences from an initial capital base of €100'000 and rebalances at the hourly frequency. At each period $t$, if $a_t \neq 0$, the strategy allocates a fixed fraction $w$ of current portfolio value $V_t$ to the position (default: $w = 10$). The notional position size in euros is therefore:

$$N_t = w \cdot V_t$$

As electricity positions are naturally expressed in energy units, the framework converts the euro notional into traded volume (MWh) using the day-ahead price $D_t$ as a sizing reference. Specifically, the traded volume is:

$$q_t = \frac{w \cdot V_t}{\max\left(|D_t|, \underline{D}\right)}$$

where $\underline{D}$ denotes a minimum price floor employed to prevent mechanically inflated volumes during episodes of very low (or near-zero) prices. In addition, a hard cap on per-trade volume is imposed to reflect practical liquidity constraints and to avoid unrealistic leverage through unit scaling. Together, the price floor and volume cap operationalise conservative feasibility constraints commonly recommended in backtesting research to reduce sensitivity to extreme observations and microstructural artefacts (De Prado, 2018, pp. 115–122).

Trade-level P&L is computed by applying the directional action to the realised spread and scaling by traded volume:

$$\mathrm{P\&L}_t^{\mathrm{gross}} = a_t \cdot S_t \cdot q_t$$

where $S_t$ denotes the realised spread at time $t$. To guard against implausibly large per-trade outcomes, gross P&L is optionally capped as a fraction of the position value $N_t$. This cap bounds the maximum gain or loss per trade to a fixed percentage of allocated notional and, combined with constant-fraction sizing, limits the portfolio impact of extreme observations in a transparent manner. Such constraints reflect the general principle that backtests should encode feasible execution and risk limits rather than assume unbounded scalability (De Prado, 2018, pp. 125–130).

Transaction costs are incorporated directly at the trade level and deducted from gross P&L. The implementation employs a fixed cost per MWh traded, so that net P&L is:

$$\mathrm{P\&L}_t^{\mathrm{net}} = \mathrm{P\&L}_t^{\mathrm{gross}} - q_t \cdot c_{\mathrm{fixed}}$$

In the baseline specification employed in this research project, the fixed transaction cost is set to $c_{\mathrm{fixed}} = 0.06$ EUR/MWh per executed trade, corresponding to 0.03 EUR/MWh for opening and 0.03 EUR/MWh for closing. When no trade is taken ($a_t = 0$) or when the spread is undefined, costs are zero. Portfolio value is updated recursively as:

$$V_{t+1} = \max\left(V_t + \mathrm{P\&L}_t^{\mathrm{net}}, 0\right)$$

with a non-negativity floor to prevent economically meaningless negative equity in the simulation.

### 3.7.2 Performance Metrics

In addition to producing an equity curve and period-by-period returns, the framework records a detailed trade log comprising the action taken, prices, spread, traded volume (MWh), gross and net P&L, and whether caps were active. Summary statistics are subsequently computed from the simulated return series, including:

- **Total Return**: Cumulative percentage change in portfolio value
- **Annualised Return**: Time-scaled return assuming continuous compounding
- **Sharpe Ratio**: Risk-adjusted return computed as the ratio of mean excess return to return volatility
- **Sortino Ratio**: Downside risk-adjusted return using only negative return observations in the denominator
- **Maximum Drawdown**: Largest peak-to-trough decline in portfolio value
- **Win Rate**: Proportion of trades yielding positive net P&L

These metrics align with standard performance measures employed in systematic trading evaluation and enable direct comparison with the statistical evaluation framework presented in Section 3.6. The emphasis throughout is that the backtest rules are fixed prior to evaluation and applied uniformly across all candidate strategies, enabling a fair out-of-sample comparison that reflects realistic execution constraints (Campbell et al., 2012, pp. 149–151; De Prado, 2018, pp. 125–130).

# 4 Results

The subsequent chapter presents the empirical results of the analysis and evaluates the performance of the proposed news-augmented modelling framework. The results are organised according to the sequential stages of the research design.

Section 4.1 provides a comprehensive overview of the finalised dataset, accompanied by a detailed exposition of descriptive statistics for the energy market variables, the news corpus, and the target variable used in the classification process. In Section 4.2, the outcomes of the feature engineering process are examined, including the results of zero-shot topic classification, embedding diagnostics, and the selection of time-decay and dimensionality-reduction parameters. Section 4.3 evaluates the predictive performance of the proposed models by comparing news-augmented specifications with two benchmarks using statistical accuracy measures and formal significance tests. Finally, Section 4.4 assesses the economic relevance of the derived signals through backtesting, reporting risk-adjusted performance metrics and drawdown characteristics under realistic transaction cost assumptions.

## 4.1 Data Overview and Descriptive Statistics

The empirical analysis is based on a combined dataset comprising 141’716 news articles and 43’798 hourly observations from the German electricity market after preprocessing, relevance filtering, and temporal alignment (Section 3.3). This dataset forms the empirical foundation for both the descriptive analysis and the subsequent evaluation of the forecasting and trading models. The joint consideration of textual and market data enables an integrated assessment of how information flows relate to short-term price dynamics.

Applying the three-class discretisation scheme (Long, Short and Neutral) specified in Section 3.4.1 yields a moderately imbalanced target distribution. Of all hourly observations, approximately 45.6% are classified as Short, 43.3% as Long, and 11.1% as Neutral (Appendix 2). This distribution indicates that economically meaningful deviations between auction and intraday prices occur frequently, whereas periods characterised by small spreads within the defined deadband are comparatively rare.

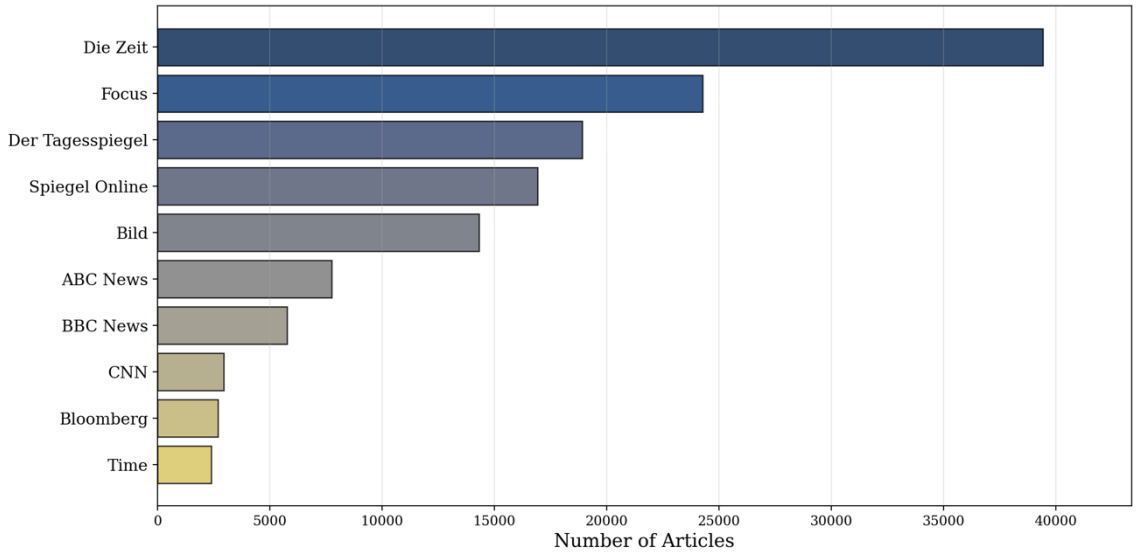


*Figure 3: Top 10 News Sources*

As illustrated in Figure 3, the corpus exhibits a high degree of concentration among primary news providers. The top five sources, comprising major German business-oriented outlets, account for 80% of all headlines. These are followed by five international sources (ranks six to ten) representing 12% of the data, while a long tail of smaller outlets provides supplementary coverage.

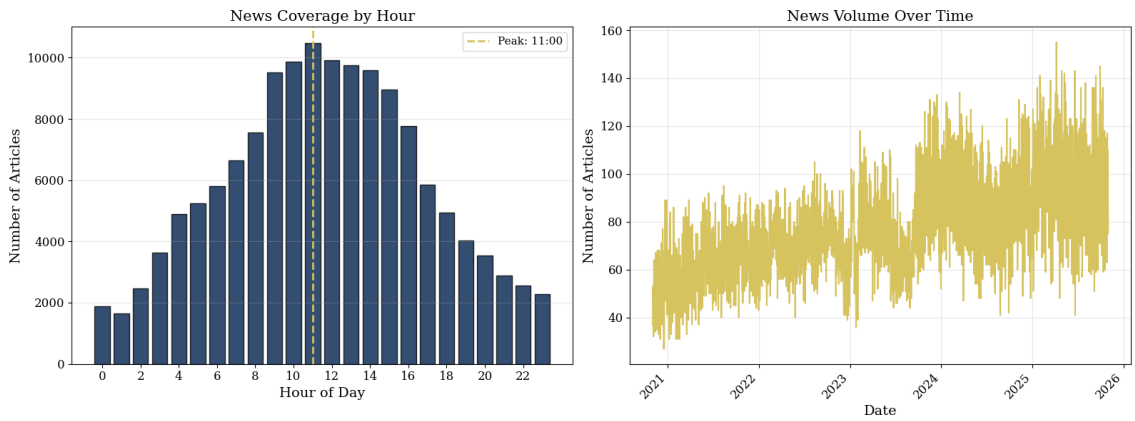


*Figure 4: News Coverage by Hour of Day*

The filtered news corpus shows pronounced temporal variation in publication intensity, as illustrated in Figure 4. News arrivals are concentrated during European business hours, with clear peaks in the late morning and early afternoon, and substantially lower activity during nighttime hours. These temporal asymmetries imply that the informational

environment faced by the model varies across hours, underscoring the need for time-aware aggregation and decay mechanisms rather than a uniform treatment of news events. In addition, the right panel of Figure 4 shows a clear upward trend in overall news volume over the sample period, reflecting an increasingly dense informational environment in energy markets, particularly in the later years of the dataset.

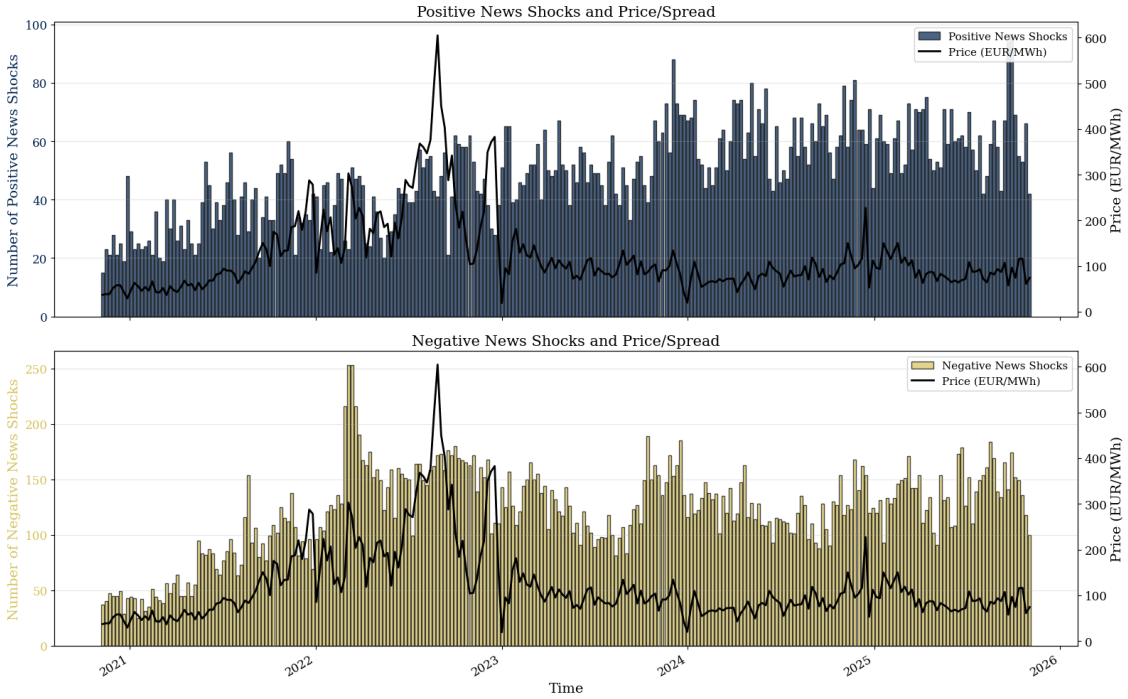


*Figure 5: News Shocks and Price/Spread*

Figure 5 provides a dual-axis illustration of aggregated positive and negative news shocks, derived from the zero-shot topic classification, alongside the corresponding price or spread series. While periods of elevated news activity sometimes coincide with notable price movements, the high level of aggregation and heterogeneity of events limit the insights that can be drawn from visual inspection alone. Accordingly, the figure is presented for descriptive purposes and serves to motivate the model-based analysis that follows.

Taken together, the descriptive statistics and visual diagnostics highlight four key characteristics of the data: concentration in news sources, heterogeneity in topical coverage, strong temporal clustering of news arrivals, and time-varying alignment between news shocks and price movements. These characteristics provide essential context for the feature engineering outcomes and model performance results discussed in Sections 4.2 to 4.4.

## 4.2 Feature Engineering Results

This section reports the empirical outcomes of the feature engineering pipeline described in Chapter 3. The results focus on the structure and composition of the news-derived features, the quality of the resulting textual representations, and the selection of key aggregation and dimensionality parameters used for downstream modelling.

The zero-shot classification framework yields a structured decomposition of the news corpus into 12 energy-relevant topic categories, while approximately 23.0% of articles are classified as non-relevant ("kein Bezug zu Energie, Wetter oder Finanzmärkten") and excluded from further aggregation (Appendix 3). This filtering outcome confirms that a substantial share of keyword-retrieved news content does not convey information with plausible relevance for electricity price dynamics. The retained articles form the basis for all subsequent topic-based and embedding-based features.

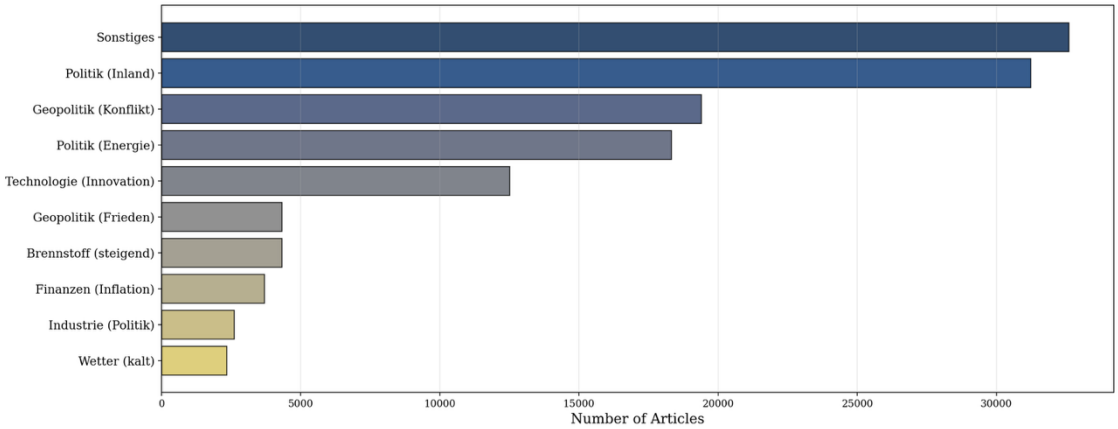


*Figure 6: Top 10 News Topic Classifications*

Figure 6 summarises the topical structure of the filtered news corpus by displaying the ten most frequent topic classifications obtained from the zero-shot framework. Topics related to domestic political developments, general political and societal news, and geopolitical conflict dominate the corpus, followed by energy-related policy topics and innovation- or technology-oriented coverage. This distribution is consistent with the broader set of political, regulatory, and macroeconomic drivers of electricity price formation discussed in Chapter 2. At the same time, the distribution highlights the heterogeneity of the incoming news flow, even after relevance filtering. A more detailed breakdown of topic frequencies is provided in Appendix 3.

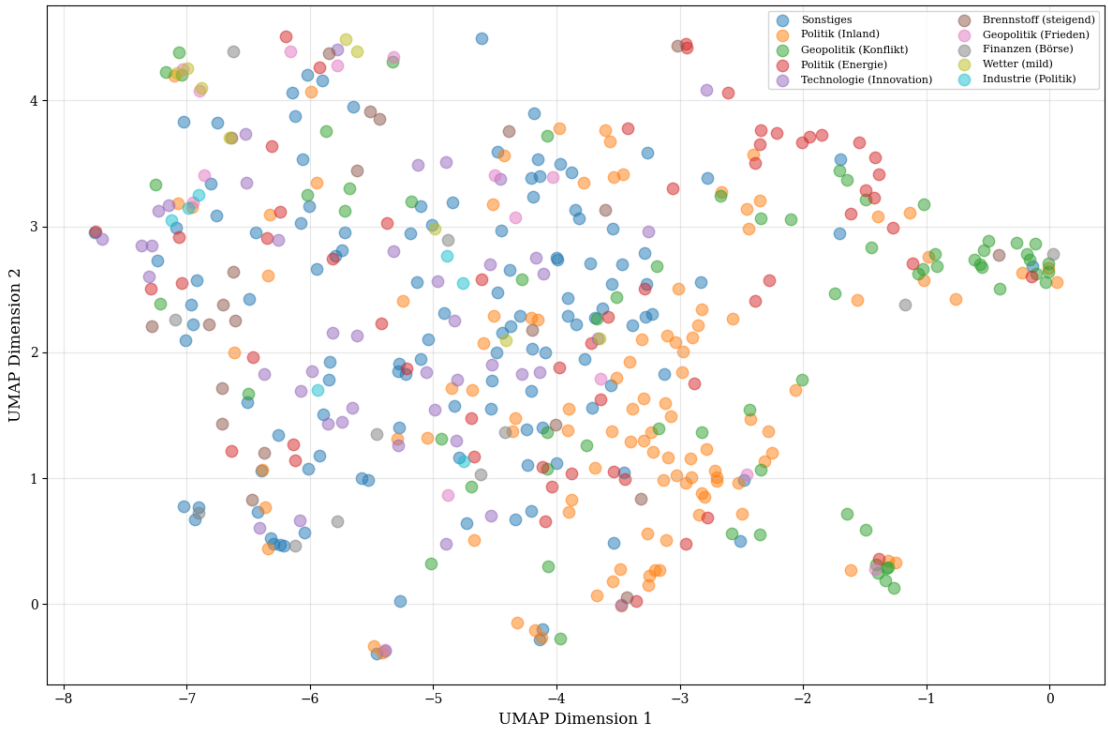


*Figure 7: News Embedding Visualisation (UMAP)*

Beyond discrete topic labels, semantic structure in the news corpus is captured through dense sentence embeddings. Figure 7 presents UMAP-based visual diagnostics of these embeddings, showing that articles assigned to certain similar topic categories tend to cluster in embedding space while remaining distinguishable from unrelated categories. The dimensionality reduction preserves the dominant semantic structure of the news corpus whilst substantially decreasing feature dimensionality. As noted in Sections 2.2.3 and 3.4.5, because the UMAP pipeline was executed in parallel and the random seed could not be frozen, minor non-determinism arises, meaning the full resulting 20-dimensional vectors for the same underlying embeddings can differ slightly across runs. These projections suggest that the embedding representations preserve meaningful semantic structure and provide a complementary, continuous representation of news content beyond the categorical topic assignments.

*Table 7: Ridge Results*

| Dataset Name | Lookback Window in h | Decay-λ | Regularization Strength (α) | Validation Accuracy | Validation Macro F1 |
|---|---|---|---|---|---|
| *dataset_lw168_dl0.01* | 168 | 0.01 | 1000.0 | 0.516 | 0.362 |
| *dataset_lw504_dl0.01* | 504 | 0.01 | 1000.0 | 0.513 | 0.354 |
| *dataset_lw168_dl0.1* | 168 | 0.10 | 1000.0 | 0.512 | 0.354 |
| *dataset_lw504_dl0.1* | 504 | 0.10 | 100.0 | 0.510 | 0.357 |
| *dataset_lw48_dl0.5* | 48 | 0.50 | 1000.0 | 0.509 | 0.358 |

The temporal aggregation of news-derived features depends critically on the choice of time-decay parameters. Following the Ridge-based screening procedure outlined in Section 3.5.2, Table 7 summarises the top-performing parameter combinations ranked by validation accuracy and macro-averaged F1 score. Across the tested configurations, models with relatively short to medium lookback windows and moderate decay rates achieve the strongest validation performance. In particular, the selected reference configuration with a 168-hour lookback window and decay parameter $\lambda = 0.01$ attains a validation accuracy of 0.516 and a macro-averaged F1 score of 0.362. This result indicates that news information retains predictive relevance over several days when incorporated with a relatively slow temporal decay. The identified parameterisation is therefore used as a benchmark for the subsequent evaluation of non-linear models.

Following dimensionality reduction of the high-dimensional sentence embeddings, the resulting 20 embedding components, together with the aggregated topic-count features, constitute the final set of news-derived inputs used in the predictive models evaluated in Sections 4.3 and 4.4.

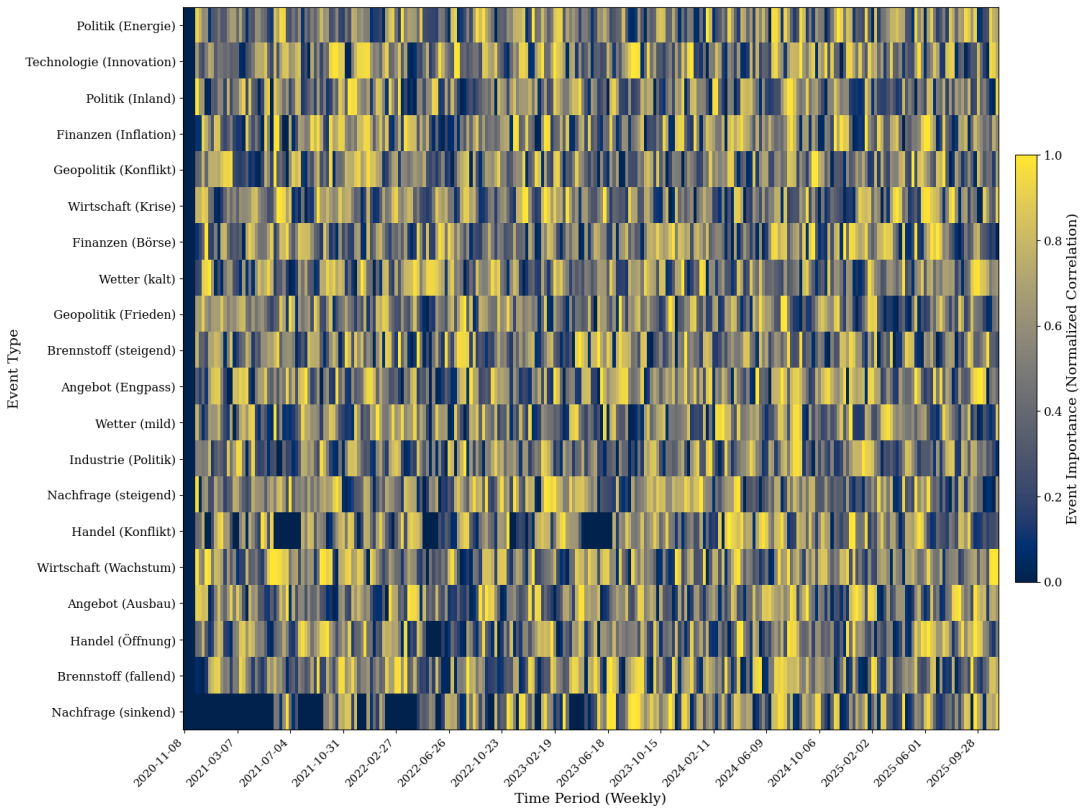


*Figure 8: Variation in Event Importance Over Time*

Figure 8 illustrates temporal variation in the relevance of different news categories through an event-importance heatmap showing the correlation between topic-specific news intensity and subsequent spread movements across time periods. The heatmap indicates that the relationship between news topics and price dynamics is not constant over time. No single news category exhibits a persistent correlation with price movements across the sample. Instead, the relative importance of different event types varies across periods. In addition, areas with near-zero event importance (dark blue regions) may partly reflect the absence or very low frequency of articles for specific topics in certain time windows, rather than a genuine lack of economic relevance.

Taken together, these results demonstrate that the feature engineering pipeline yields a structured and semantically coherent representation of news information, while also revealing substantial time variation in topical relevance. These properties motivate the use of flexible, non-linear models and selective signal construction in the performance evaluation that follows.

## 4.3 Statistical Evaluation and Predictive Performance

This section evaluates the out-of-sample predictive performance of the proposed models and assesses whether the inclusion of news-derived information leads to statistically meaningful improvements over a price-based benchmark. All results are reported on the held-out test set and focus on multi-class classification performance across Long, Neutral, and Short signals.

The news-aware XGBoost three-class model is evaluated here as the first-stage component of the proposed two-stage ensemble pipeline, in order to quantify its individual contribution when trained on the combined set of market-based and news-derived features. Based on the training and validation results reported in Table 7, the configuration with a 168-hour lookback window and a decay parameter of $\lambda = 0.01$ is selected for out-of-sample evaluation. On the held-out test set, this model achieves a test accuracy of 50.5%, a macro-averaged F1 score of 0.410, and a macro-averaged AUC of 0.61, with detailed results reported in Appendix 4. Learning curve diagnostics (Appendix 5) further indicate a gradual improvement in validation performance as the training sample expands, suggesting that additional data could lead to higher out-of-sample accuracy. These results indicate that the model captures economically relevant information beyond random

classification in a setting characterised by class imbalance and noisy short-term price dynamics.

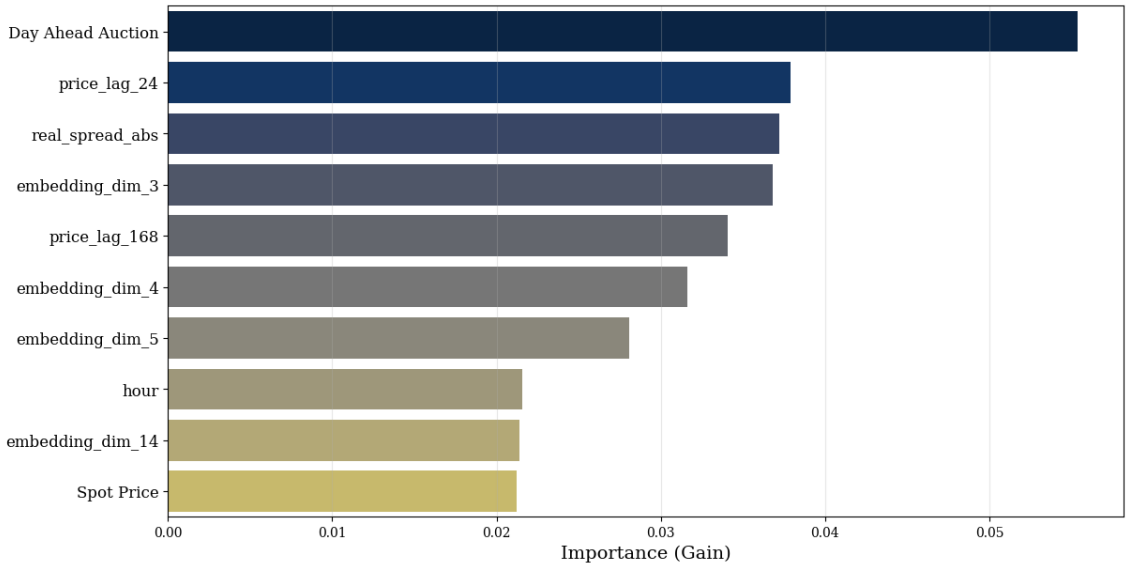


*Figure 9: XGBoost Top 10 Features (Gain)*

Figure 9 shows the top ten features by gain-based importance in the XGBoost classifier. Market-based variables, such as the day-ahead auction price and price lags, dominate the ranking, reflecting strong persistence in electricity prices. At the same time, several news-derived embedding dimensions appear among the most influential features, indicating that textual information provides incremental predictive value beyond historical prices alone. Notably, none of the time-decayed topic-count features appear among the top ten by gain, suggesting that the model extracts more predictive signal from the dense news embeddings than from the time-decayed topic-count representation.

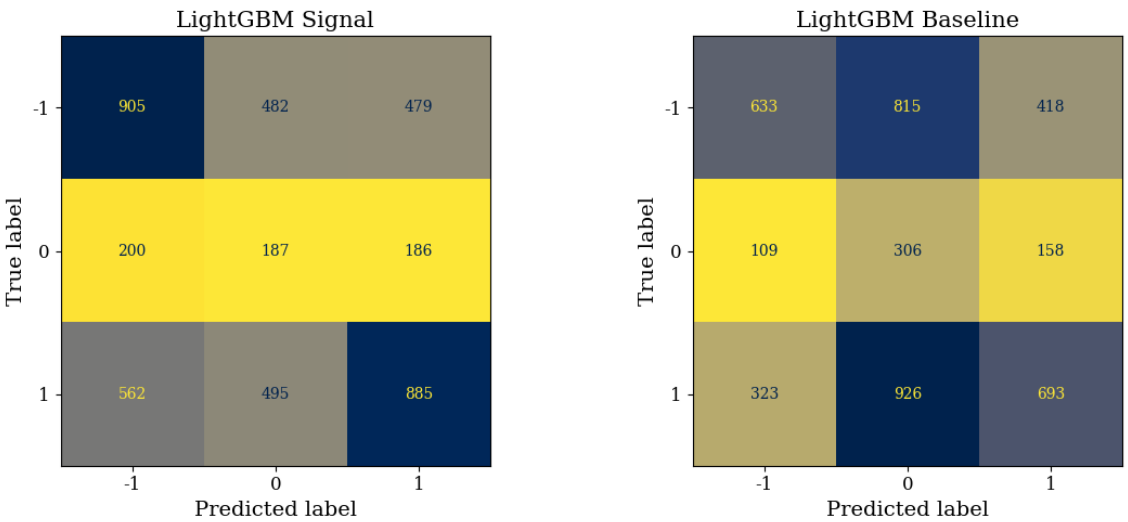


*Figure 10: Confusion Matrices for Signal and Baseline Model*

*Table 8: LightGBM Classification Report (Test Set)*

| Class | Model | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|---|
| Short (-1) | Signal | 0.54 | 0.48 | 0.51 | 1866 |
| | Baseline | 0.59 | 0.34 | 0.43 | 1866 |
| Neutral (0) | Signal | 0.16 | 0.33 | 0.22 | 573 |
| | Baseline | 0.15 | 0.53 | 0.23 | 573 |
| Long (1) | Signal | 0.57 | 0.46 | 0.51 | 1942 |
| | Baseline | 0.55 | 0.36 | 0.43 | 1942 |
| Macro Avg | Signal | 0.42 | 0.42 | 0.41 | 4381 |
| | Baseline | 0.43 | 0.41 | 0.37 | 4381 |

| | Signal | Baseline |
|---|---|---|
| **Accuracy** | 0.451 | 0.373 |
| **Macro AUC** | 0.600 | 0.594 |

Following the two-stage architecture described in Section 3.5, the calibrated XGBoost class probabilities serve as inputs to the LightGBM signal model. As reported in Table 8, the signal model achieves a test accuracy of 45.1% and a macro-averaged F1 score of 0.41, outperforming the corresponding LightGBM baseline model, which relies solely on price-based inputs and attains an accuracy of 37.3% and a macro- averaged F1 score of 0.37. While overall accuracy is lower than that of the standalone XGBoost classifier, the signal model exhibits a more balanced performance across classes, reflected in its higher macro-F1 score. Figure 10 presents the corresponding confusion matrices for the signal and baseline models, illustrating improved discrimination between Long and Short outcomes under the signal specification, while misclassification remains most pronounced for the Neutral class. The signal model also attains a slightly higher macro-averaged AUC than the baseline model, indicating improved class discrimination across probability thresholds; the corresponding Receiver Operating Characteristic curves are shown in Appendix 6.

The performance difference between the signal model and the baseline model is formally assessed using McNemar's test for paired classification outcomes (Appendix 7). The resulting test statistic of $\chi^2 = 113.89$ with a corresponding p-value below 0.001 rejects the null hypothesis of equal predictive accuracy. Complementary bootstrap analysis of test-set accuracy further supports this finding, as the 95% confidence intervals of the two models do not overlap, providing robust evidence that the signal model delivers statistically significant improvements over the baseline.

An analysis of feature importance reveals a clear distinction between the two LightGBM specifications. In the signal model, the most influential features are the XGBoost prediction probabilities, indicating that the upstream classifier effectively condenses information from the news-derived inputs into a small number of high-value signals. In contrast, the baseline model relies primarily on price-based variables, such as the day-ahead auction price and absolute spread measures, reflecting its dependence on short-term price persistence rather than external information. The feature importance rankings for both models are reported in Appendix 8 and Appendix 9.

In summary, the empirical results show that models incorporating news-derived information achieve higher predictive performance than the price-only benchmark across multiple evaluation metrics. The signal model consistently attains higher macro-averaged F1 scores and AUC values and outperforms the baseline model in terms of predictive accuracy, with the difference confirmed as statistically significant by McNemar's test and bootstrap confidence intervals. At the same time, absolute performance levels remain moderate, reflecting the noisy and volatile nature of short-term electricity price movements. These results complete the statistical evaluation of the predictive models.

## 4.4 Economic Evaluation and Backtesting Performance

This section evaluates the economic performance of the proposed trading strategies based on the predictive signals examined in Section 4.3. A stylised portfolio backtest is conducted over the test period to assess whether the statistical improvements of the news-based signal model translate into superior trading outcomes relative to a price-based baseline and a naïve long-only benchmark strategy. The portfolio simulation follows the specification detailed in Section 3.7.1, with all strategies evaluated under identical execution and cost assumptions to ensure comparability.

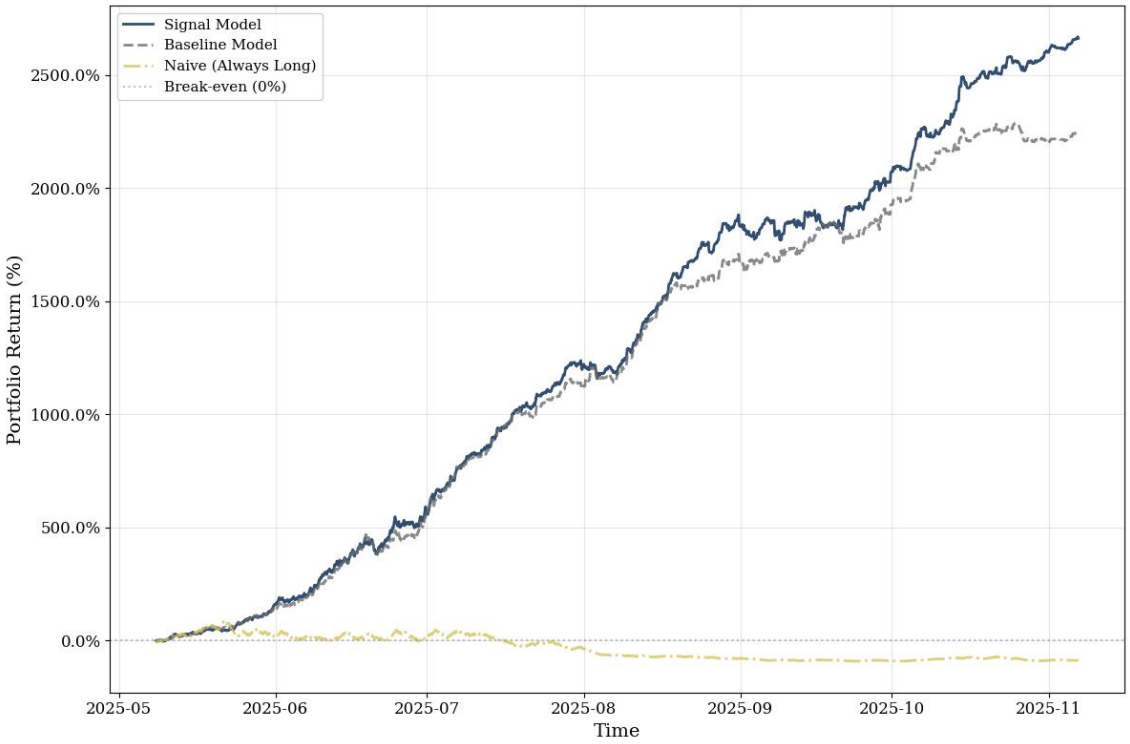


*Figure 11: Portfolio Equity Curve - Test Period*

Figure 11 displays the cumulative portfolio returns and corresponding drawdowns for the signal model, the baseline model, and a naïve benchmark strategy that remains continuously long. Both model-based strategies generate substantial cumulative gains over the test period, while the naïve strategy experiences a severe and persistent drawdown, ultimately resulting in large capital losses. The signal model achieves the highest terminal portfolio value, reaching approximately €2’76 million, compared with €2’35 million for the baseline model. In contrast, the naïve strategy ends the sample with a final portfolio value of roughly €13’000, reflecting the pronounced downside risk of an unconditional long exposure in the German power market.

*Table 9: Comparative Portfolio Performance and Risk-Adjusted Metrics*

| **Strategy** | *Signal Model* | *Baseline Model* | *Naïve (Always Long)* |
|---|---|---|---|
| **Final Value (EUR)** | 2’760’445 | 2’349’169 | 13’191 |
| **Total Return (%) *** | 2’660.45 | 2’249.169296 | −86.81 |
| **Annualised Return (%) *** | 75’985.23 | 55’006.49 | −98.26 |
| **Sharpe Ratio *** | 1’552.47 | 1’059.35 | −1.22 |
| **Sortino Ratio** | 2’197.08 | 1’561.20 | −3.18 |
| **Max Drawdown (%)** | −4.04 | −3.99 | −95.11 |
| **Win Rate (%)** | 61.36 | 62.17 | 49.62 |
| **Total Trades** | 3’217 | 2’334 | 4’381 |
| **Total Costs (EUR)** | 313’329 | 226’146 | 174’609 |

** High Returns & Sharpe Ratio due to simplified trading assumptions and limitations (Section 4.2 & 5.3)*

Table 9 summarises the key performance metrics for all three strategies. The signal model records a total return of 2'660%, outperforming the baseline model's 2'249% return. Risk-adjusted performance is substantially higher for the signal model, with an annualised Sharpe ratio of 1'552 and a Sortino ratio of 2'197, compared to 1'059 and 1'561, respectively, for the baseline strategy. Both models achieve win rates above 60%, while the naïve strategy remains close to a coin-flip outcome.

The signal model executes a larger number of trades than the baseline strategy, reflecting its more active use of information, and incurs higher absolute transaction costs. Nevertheless, these costs are more than offset by the higher gross returns achieved. The naïve strategy, despite executing the largest number of trades, generates strongly negative performance and exhibits the weakest risk-adjusted metrics.

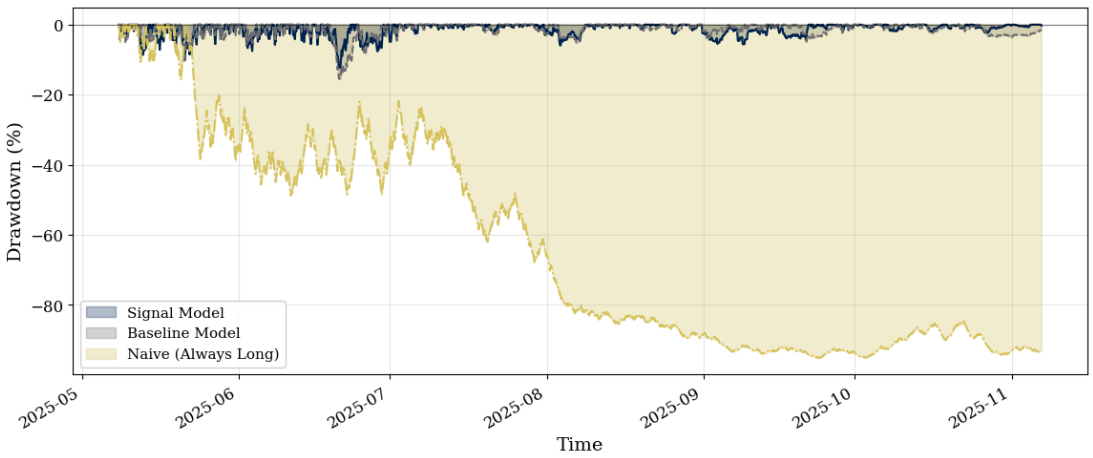


*Figure 12: Drawdown (% of Peak)*

In terms of trade-level outcomes, both model-based strategies achieve win rates above 60%, as reported in Table 9, indicating that profitable trades occur more frequently than losing trades under both specifications. The baseline model records a slightly higher win rate than the signal model; however, this difference is accompanied by lower cumulative returns and weaker risk-adjusted performance.

Drawdown behaviour further differentiates the strategies. As summarised in Table 9 and illustrated in Figure 12, the signal and baseline models both experience relatively shallow maximum drawdowns of approximately −4%, with drawdowns remaining contained throughout the test period. In contrast, the naïve always-long strategy exhibits severe and persistent drawdowns, reaching a maximum drawdown of over −95%, with limited

recovery. The drawdown trajectories in Figure 12 show that losses for the model-based strategies are short-lived and recover quickly, whereas the naïve strategy remains exposed to prolonged downside risk.

In summary, the backtesting results show that both model-based strategies generate substantial profits under identical execution and cost assumptions, whereas the naïve always-long strategy performs poorly and is exposed to severe downside risk. The signal model achieves the highest terminal portfolio value and the strongest risk-adjusted performance, despite incurring higher transaction costs due to more frequent trading. Drawdown analysis further confirms that losses for the model-based strategies remain limited and recover quickly, in contrast to the persistent drawdowns observed for the naïve benchmark. Together, these results conclude the empirical evaluation of economic performance and provide the basis for the discussion of robustness, limitations, and broader implications in the subsequent chapter.

# 5 Discussion

This chapter synthesises and critically evaluates the empirical findings presented in Chapter 4, situating the results within the broader academic literature on news-based forecasting and trading in energy markets. The discussion proceeds in five parts: Section 5.1 interprets the statistical and economic performance of the proposed models; Section 5.2 positions these findings relative to prior research reviewed in Chapter 2; Section 5.3 reflects on methodological robustness and limitations; Section 5.4 distils the core findings and articulates the study's contribution; and Section 5.5 outlines practical implications and directions for future research.

## 5.1 Interpretation of the Model Performance

This section interprets the empirical performance of the proposed modelling framework based on the statistical and economic results presented in Sections 4.3 and 4.4. The focus lies on understanding how the integration of news-derived information affects classification quality, class balance, and downstream trading performance relative to price-based and naïve long-only benchmarks.

From a classification perspective, the two-stage modelling approach demonstrates consistent improvements across multiple evaluation dimensions. The upstream XGBoost classifier exhibits performance clearly above random classification across accuracy, macro-F1, and AUC metrics (Section 4.3), despite pronounced class imbalance in the target distribution, where Neutral outcomes account for only a small fraction of observations. This indicates the presence of non-trivial predictive structure beyond historical price dynamics alone. Importantly, the use of probability-based decision rules rather than naïve argmax classification improves class balance, particularly by increasing Neutral recall while maintaining comparable predictive quality for Long and Short outcomes. This behaviour aligns with established findings on probability calibration in gradient-boosted ensembles, where raw model outputs frequently exhibit systematic over- or underconfidence that degrades downstream decision quality (Niculescu-Mizil & Caruana, n.d., pp. 1–2).

The downstream LightGBM signal model further refines these predictions by embedding the calibrated XGBoost class probabilities into a second-stage decision framework.

Relative to the LightGBM baseline model, the signal model achieves higher overall accuracy and a markedly stronger macro-averaged F1 score on the held-out test set (Section 4.3; Table 8). More importantly, these gains are driven by improved balance across classes rather than by dominance in a single outcome category. While the baseline model exhibits a tendency to over-assign Neutral positions, the signal model achieves more even recall across Long, Neutral, and Short outcomes, as illustrated by the corresponding confusion matrices (Section 4.3; Figure 10). This suggests that the news-derived features add complementary information beyond market variables, improving class separability and reducing the baseline model's tendency to default to Neutral trades, thereby supporting more balanced predictions across Long, Neutral, and Short.

Discrimination performance across probability thresholds shows a similar ordering. The signal model records a macro-averaged AUC of 0.600, marginally exceeding the baseline model's AUC of 0.594 (Section 4.3; Table 8). Whilst the absolute difference is modest, it is consistent with the observed gains in macro-F1 and reflects improved separation between Long, Neutral, and Short outcomes when news-derived information is incorporated. The performance difference is formally validated by McNemar's test (Appendix 7), which rejects the null hypothesis of equal predictive accuracy at the 1% significance level. These results indicate that the textual features contribute incremental information beyond what is contained in historical prices and temporal patterns alone.

The economic relevance of these classification improvements becomes evident in the portfolio backtesting results presented in Section 4.4. Under identical execution and cost assumptions, both model-based strategies generate substantial cumulative gains over the test period, while the naïve benchmark strategy that remains continuously long performs poorly. The signal model achieves the highest terminal portfolio value and total return, outperforming the price-based baseline, while the naïve strategy suffers severe capital erosion and persistent drawdowns (Section 4.4; Table 9). This contrast illustrates that directional timing, rather than unconditional exposure, is essential in the German power market and that the predictive models capture economically actionable information. Risk-adjusted performance metrics reinforce this interpretation. The signal model exhibits superior Sharpe and Sortino ratios relative to the baseline strategy, indicating a more favourable balance between returns and downside risk (Section 4.4; Table 9). At the same time, both model-based strategies maintain shallow and well-contained maximum drawdowns of around 4%, with rapid recovery following adverse periods. In contrast, the naïve

strategy experiences extreme and prolonged drawdowns exceeding 90%, underscoring its lack of robustness in volatile electricity markets (Section 4.4; Figure 12).

At the trade level, both model-based strategies achieve win rates above 60%, whereas the naïve strategy remains close to a coin-flip outcome (49.6%). Although the signal model trades more actively and incurs higher absolute transaction costs, these costs are more than offset by higher gross profitability, resulting in superior net performance (Section 4.4; Table 9). This pattern suggests that the news-derived signals not only improve directional accuracy but also support more effective trade timing and capital deployment.

Overall, the performance metrics consistently indicate that the inclusion of news-derived signals improves predictive balance, class discrimination, and economic outcomes when integrated into a structured, two-stage modelling framework. The gains are observable at the level of statistical accuracy, probabilistic discrimination, and realised trading performance, providing a coherent empirical picture across model layers and evaluation domains.

## 5.2 Positioning of Findings Within the Literature

This section situates the empirical findings of this research project within the existing literature on news-based forecasting and trading in electricity and other financial markets, as reviewed in Sections 2.3.1 and 2.3.2. The objective is not to restate empirical performance, but to assess how the results confirm, extend, or qualify prior findings and to clarify the specific contribution of the proposed modelling framework.

A first point of alignment with the literature concerns the incremental value of textual information for price forecasting. Prior studies consistently document that news-derived features improve performance relative to traditional models relying solely on historical prices, load, and weather variables (Bai et al., 2024, pp. 5–6; Bi et al., 2025, pp. 7–9). While the fundamental drivers identified in the broader forecasting literature – such as weather, demand patterns, and physical system constraints – remain central to price formation (Hong et al., 2020, pp. 377–378), the results of this research project show that the inclusion of energy-related news yields statistically significant improvements in predictive performance beyond these standard inputs. Importantly, these gains persist even when the baseline benchmark model already captures strong price persistence and intraday

structure, reinforcing the interpretation of news as a complementary rather than substitutive source of information.

Beyond confirming earlier forecasting results, this study directly addresses a limitation highlighted in Section 2.3.2, namely, the gap between improved predictive accuracy and economically actionable trading signals. While much of the existing literature evaluates performance at the forecasting level (Section 2.3.1), only a limited number of studies explicitly embed textual signals into disciplined trading frameworks that account for execution rules, class imbalance, and asymmetric payoffs. By separating forecast generation from signal construction within a two-stage modelling framework, this research project operationalises the conceptual distinction discussed in Section 2.3.2 and demonstrates that news-augmented forecasts can be transformed into economically meaningful trading decisions. This interpretation is consistent with the findings of Chen et al. (2025, pp. 2–5), who emphasise that realised performance in electricity markets depends critically on market constraints and execution design rather than predictive error alone.

The findings also complement recent work employing deep learning architectures for news-based forecasting and trading. Studies such as Chakraborty et al. (2024) and Bi et al. (2025) highlight the ability of transformer-based representations to capture complex semantic information and to improve prediction during periods of heightened uncertainty. While these approaches often rely on end-to-end neural architectures, the present research project adopts a different modelling philosophy by combining transformer-based embeddings with gradient-boosted decision trees and an explicit decision layer. The observed performance suggests that economically relevant information contained in news can be effectively distilled into lower-dimensional representations and exploited by non-linear but structured models, supporting the argument in Section 2.3.2 that signal design is as important as raw forecasting power.

An additional contribution relative to the trading-focused literature lies in the explicit consideration of transaction costs and market-specific constraints. As discussed in Section 2.3.2, the economic value of NLP-based forecasts depends critically on execution discipline and market frictions. The results of this research project show that the predictive value of news-derived signals persists under conservative transaction cost assumptions and fixed position sizing rules. This is particularly relevant for electricity markets, where frequent trading, auction-based price formation, and intraday adjustments impose

structural constraints on real-world implementation (Müsgens & Bade, 2024, pp. 3–4; Edwards, 2023, pp. 27–28). While these results are conditional on the stylised backtesting assumptions and the specific market environment defined in this study, the persistence of performance gains under such constraints provides a preliminary indication of economic relevance and helps bridge the gap toward more execution-oriented research.

At the same time, the results also qualify some expectations set by the literature. While several studies report substantial performance improvements from news-based models, the gains observed in this research are incremental rather than transformative. This outcome is consistent with the interpretation, emphasised in Section 3.4.4, that electricity markets are already highly information-rich, with prices rapidly incorporating publicly available signals (Weron, 2014, pp. 1031–1032). In such an environment, news-derived information functions primarily as a filter that refines directional timing and reduces noise, rather than as a dominant driver of price dynamics.

Taken together, the findings of this research project confirm the relevance of textual information for electricity price forecasting (Section 2.3.1), extend the literature by demonstrating its economic actionability within a structured trading framework (Section 2.3.2), and qualify expectations regarding the magnitude of achievable gains. By explicitly linking news-based forecasting to signal construction and backtesting, the study contributes to a more integrated understanding of how NLP techniques can be applied in practice to short-term power market trading.

## 5.3 Methodological Robustness and Limitations

This section reflects critically on the methodological robustness of the proposed framework and discusses the key limitations that condition the interpretation and generalisability of the empirical results. The objective is not to question the internal validity of the findings, but to clarify the assumptions under which they hold and to identify dimensions where further refinement would be required for real-world deployment.

A first set of limitations arises from the construction of the news dataset. Although the corpus comprises many observations, the initial retrieval of news articles relies on keyword-based filtering prior to NLP processing. While this approach is necessary to ensure topical relevance and computational feasibility, it introduces potential selection bias by

excluding articles that may affect electricity prices indirectly or through less explicit terminology. In addition, news coverage is uneven across time and topics, with clustering around major events and reduced publication intensity during off-peak hours (Section 4.1; Figure 4). Although the subsequent zero-shot relevance classification and time-decay aggregation mitigate these effects, the informational environment faced by the model remains heterogeneous, which may influence signal strength across periods.

The zero-shot topic classification framework itself introduces further limitations. Topic hypotheses are defined ex ante to ensure interpretability and domain alignment, but classification outcomes necessarily depend on the quality and completeness of these hypotheses. Ambiguous or multi-faceted news events may be assigned imperfectly, particularly when headlines span several thematic dimensions simultaneously. In addition, approximately 23% of the retrieved news articles are classified into a residual "Sonstiges" (no energy relevance) category by the zero-shot framework. Rather than performing a deeper post hoc analysis of these articles to refine topic hypotheses or expand the classification scheme, they are excluded from subsequent feature construction. While this filtering step improves focus and reduces noise, it also implies that potentially relevant information may be discarded at an early stage of the pipeline. Moreover, the zero-shot model is not fine-tuned on domain-specific energy texts, implying that classification accuracy relies on general language understanding rather than task-specific optimisation. These constraints are consistent with the broader limitations of unsupervised and zero-shot approaches discussed in Section 2.2.3 as well as 3.4.5 and imply that some semantic nuance may be lost at the topic-assignment stage.

A central methodological limitation concerns the use of dimensionality reduction techniques for embedding compression. Transformer-based sentence embeddings provide rich semantic representations, but their high dimensionality necessitates compression before integration into downstream models. In this research project, UMAP is employed for this purpose due to its ability to preserve local and global structure. However, as highlighted in Sections 2.2.3 and 3.4.5, UMAP is inherently slightly sensitive to random initialisation and hyperparameter choices, resulting in non-deterministic projections across runs. While overall performance patterns remain stable, exact numerical results cannot be reproduced deterministically without fixing random seeds and configurations. This limitation complicates strict reproducibility and underscores the trade-off between representational richness and methodological stability.

The modelling architecture and hyperparameter selection introduce additional sources of sensitivity. Although gradient-boosted decision trees are well suited to handling noisy, non-linear features, model performance depends on choices regarding lookback windows, decay parameters, and probability thresholds. These parameters are selected through cross-validation and out-of-sample testing, reducing the risk of overfitting, but do not eliminate it entirely. In particular, the optimal configuration may vary across market regimes, suggesting that static parameter choices could underperform in environments characterised by structural change.

Limitations also arise from the design of the trading signal and backtesting framework. The portfolio simulation adopts stylised assumptions, including fixed position sizing, constant transaction costs, and immediate execution at observed prices. In addition, trade execution is based on aggregated price measures, relying on mean hourly prices rather than explicitly modelling intra-hour volatility or liquidity constraints characteristic of 15-minute intraday trading. As a result, the simulation abstracts from intraday price dispersion and execution uncertainty, which may materially affect realised entry and exit prices in practice. Market impact, liquidity constraints, bid–ask spreads, and dynamic risk management mechanisms are likewise not modelled. While these simplifications allow for controlled comparison across strategies and align with the conceptual framework outlined in Section 2.3.2, they imply that realised performance in live trading environments may differ materially from simulated outcomes, particularly in periods of heightened intraday volatility.

The generalisability of the results is further constrained by market- and sample-specific factors. The analysis focuses exclusively on the German power market over a period that includes exceptional events such as energy supply disruptions and policy interventions. While this enhances the relevance of news-based signals, it also implies that model performance may be regime-dependent. As noted by Bi et al. (2025), large language model–based representations can be sensitive to shifts in market structure and vocabulary, raising concerns about stability when applied to future periods or other power markets without recalibration.

Finally, while statistical robustness is supported through out-of-sample testing, McNemar's test, and bootstrap confidence intervals, economic robustness is evaluated under a single set of trading assumptions. Alternative cost structures, execution

frequencies, or risk constraints are not explored. Consequently, the reported economic performance should be interpreted as conditional on the chosen simulation design rather than as a universal performance benchmark.

Taken together, these considerations indicate that the proposed framework is methodologically robust within its defined scope but subject to important limitations related to data construction, representation learning, reproducibility, and execution realism. Acknowledging these constraints reinforces the interpretation of the results as a structured proof of concept and motivates the broader implications and avenues for future research discussed in Section 5.5.

## 5.4 Core Findings and Contribution

This section consolidates the principal findings of the research project and articulates the study's contribution to the academic literature on news-based forecasting and trading in electricity markets. The discussion synthesises the statistical and economic evidence presented in Chapter 4, drawing upon the theoretical and empirical foundations established in Chapter 2 to clarify how the results advance current understanding of textual information processing in short-term power markets.

The empirical analysis yields three core findings. First, the inclusion of news-derived features produces statistically significant improvements in predictive performance relative to models relying solely on historical prices and temporal patterns. The signal model achieves higher accuracy, macro-averaged F1 scores, and AUC values than the baseline specification, with the performance differential confirmed by McNemar's test at the 1% significance level (Section 4.3). This result aligns with prior findings in commodity and energy markets, where news-augmented models have consistently outperformed price-only benchmarks, particularly during periods characterised by heightened uncertainty and information-driven volatility (Smales, 2014, pp. 285–286; Li et al., 2021, pp. 3–4; Guan et al., 2022, pp. 5–10; Bai et al., 2024, pp. 5–6). The present study extends this evidence to the German electricity market, demonstrating that textual information retains predictive relevance even in a setting where prices already incorporate rich physical and temporal structure.

Second, the statistical improvements translate into economically meaningful trading outcomes under simplified execution assumptions. The signal model achieves superior cumulative returns, Sharpe ratios, and Sortino ratios relative to both the price-based baseline and a naïve long-only benchmark, whilst maintaining contained maximum drawdowns (Section 4.4). This finding addresses a recurrent limitation in the forecasting literature, where predictive accuracy is frequently evaluated in isolation from implementability constraints (J. Chen et al., 2025, pp. 18–21; Kirtac & Germano, 2024, pp. 7–8). The persistence of performance gains under moderate cost assumptions strengthens confidence in the economic relevance of the proposed approach and differentiates it from purely statistical contributions.

Third, the two-stage stacked ensemble architecture seems effective in distilling high-dimensional textual representations into actionable trading signals. Feature importance analysis reveals that the meta-learner assigns the greatest weight to the upstream probability estimates, indicating that the XGBoost classifier successfully condenses semantic information from news into a compact set of decision-relevant inputs (Section 4.3; Appendix 8). This architectural choice responds to concerns raised in the literature regarding the opacity and computational cost of end-to-end deep learning approaches (Chakraborty et al., 2024, pp. 24–25; Das, 2014, pp. 106–107). By combining transformer-based embeddings with interpretable gradient-boosted models, the framework balances representational richness against transparency and computational feasibility, which is a trade-off of particular relevance in electricity markets, where rapid information processing and discretionary oversight remain operationally important (Müsgens & Bade, 2024, pp. 153–154).

The study contributes to the literature in three respects. Methodologically, it demonstrates that the conceptual distinction between forecasting and trading, which was emphasised by Kirtac and Germano (2024, pp. 3–4) and Chen et al. (2025, pp. 1–3), can be operationalised through a modular pipeline that separates feature engineering, probabilistic classification, and signal construction. Empirically, the study extends the evidence base for news-augmented electricity price forecasting beyond the datasets and market settings examined in recent contributions (Bai et al., 2024; Bi et al., 2025), providing out-of-sample validation over a five-year period that encompasses both crisis and stabilisation regimes in the German power market. Conceptually, the results reinforce the view that textual information functions as a complementary filter that refines directional timing rather than

as a dominant predictor of price levels, which is consistent with Tetlock's (2007, pp. 1149–1151) characterisation of news impact as subject to rapid decay as information is absorbed into market prices.

Consequently, the primary research question of this research project can be answered as follows: energy-related news can be transformed into statistically and economically significant trading signals for the German power market within the confines of the proposed dual NLP-based modelling framework and the stated backtesting assumptions. The signal model outperforms the price-based baseline and naïve benchmarks across both classification metrics and risk-adjusted return measures, providing affirmative evidence for the incremental value of textual information in short-term electricity trading. However, the results should be interpreted as an empirical proof of concept rather than as a ready-to-deploy trading system. As discussed in Section 5.3, limitations related to data construction, dimensionality reduction, and execution realism imply that further methodological refinement, robustness testing, and implementation considerations are required before such a model could be considered suitable for real-world deployment. Within these boundaries, the findings offer a structured demonstration that modern NLP techniques can extract economically actionable information from the continuous flow of energy-related news.

## 5.5 Practical Implications & Future Research

The findings of this research project carry implications for market practitioners operating in day-ahead and intraday electricity markets. For proprietary trading desks and portfolio managers, the demonstrated value of news-derived features suggests that investment in NLP infrastructure, including news aggregation pipelines, transformer-based embedding models, and probability-calibrated classifiers, may yield incremental improvements over strategies relying solely on price-based technical indicators. For utilities and large industrial consumers engaged in hedging activities, the framework offers potential enhancements to procurement optimisation by providing early signals of spread direction that could inform the timing of forward contract execution. Nevertheless, practitioners should recognise that live deployment would require careful calibration to prevailing liquidity conditions and market microstructure, and that discretionary oversight remains essential

for responding to unexpected events not fully captured by automated systems (Müsgens & Bade, 2024, pp. 153–154; Edwards, 2023, pp. 241–242).

Several avenues for future research emerge from this study. Methodological extensions could explore domain-specific language models fine-tuned on energy market texts, which may capture sector-specific terminology more effectively than general-purpose transformers (Kaplan et al., 2023, p. 1; Loughran & McDonald, 2020, pp. 4–5). The temporal aggregation of news features could be refined through adaptive decay mechanisms that adjust dynamically to market volatility regimes. Future work could also extend the geographical scope to other European power markets or cross-border trading settings (Frauendorfer et al., 2018, pp. 2–3), and integrate alternative textual sources such as social media discourse, regulatory filings, or grid operator communications. Finally, real-time implementation studies accounting for execution latency, market impact, and dynamic position management would address the gap between retrospective backtesting and live market deployment (De Prado, 2018, pp. 115–122).

# 6 Conclusion

This research project examined whether textual information contained in energy-related news can be systematically transformed into statistically and economically meaningful trading signals for the German electricity market. Motivated by the growing availability of unstructured textual data and recent advances in NLP, the study focused on short-term price dynamics in a market characterised by high volatility, strong intraday structure, and rapid information absorption.

To address this objective, the thesis developed and evaluated a modular modelling framework that integrates news headlines with traditional market variables. Textual information was processed through a structured NLP pipeline combining zero-shot relevance filtering and topic classification with transformer-based sentence embeddings. This design enabled the extraction of semantically rich but interpretable features without relying on domain-specific supervised labelling, thereby maintaining flexibility and scalability. The resulting news-derived features were integrated with price-based inputs using a two-stage modelling architecture that separates forecast generation from trading signal construction.

At the forecasting layer, gradient-boosted decision trees in the form of XGBoost were employed to map combined market and news features to probabilistic class forecasts. This choice reflects a deliberate balance between non-linear modelling capacity and robustness to noisy, high-dimensional inputs. Rather than treating the classifier output as a direct trading signal, the probabilistic forecasts were subsequently embedded into a second-stage LightGBM model designed specifically for signal construction. This stacking approach allowed the downstream model to exploit the information content of the XGBoost probability estimates while incorporating additional price-based context, thereby operationalising the conceptual distinction between prediction and decision-making.

The empirical analysis demonstrates that incorporating news-derived information yields consistent improvements over the price-only benchmark across both statistical and economic dimensions. At the forecasting level, models that combine zero-shot filtered news embeddings with price features exhibit improved class balance and discrimination, indicating that textual information contributes incremental signals beyond historical prices and temporal patterns alone. At the trading level, these improvements persist under conservative transaction-cost assumptions and disciplined signal construction, yielding materially different portfolio outcomes relative to both a price-based baseline and a naïve

long-only benchmark strategy. Importantly, the results show that the economic relevance of news does not stem from amplifying exposure, but from refining directional timing and reducing exposure during unfavourable regimes.

Beyond the empirical findings themselves, the study contributes to the literature in several respects. First, it reinforces prior evidence that textual information can enhance short-term electricity price forecasting by capturing expectation-driven and event-based signals not directly observable in physical system data. Second, it extends this literature by demonstrating that such improvements remain economically meaningful when forecasts are transformed into trading signals using a structured, execution-aware framework. Third, the combination of zero-shot classification, XGBoost forecasting, and LightGBM signal construction illustrates how modern NLP techniques can be integrated into established machine-learning pipelines without resorting to opaque end-to-end architectures, thereby preserving interpretability and modularity.

At the same time, the findings should be interpreted within the methodological boundaries established in the preceding chapters. As discussed in detail in Chapter 5, the results are conditional on several design choices related to data construction, representation learning, dimensionality reduction, and backtesting assumptions. In particular, reliance on zero-shot classification makes the formulation of topic hypotheses sensitive, while the use of non-deterministic dimensionality-reduction techniques, such as UMAP, introduces an additional source of variability, as embedding projections may differ across runs depending on initialisation and hyperparameter settings. Combined with the probabilistic stacking of XGBoost and LightGBM models, this implies that exact numerical outcomes are not strictly reproducible without fixing all random seeds and configurations. These constraints do not invalidate the empirical results, but they frame them as a structured proof of concept rather than as evidence of immediate deployability in live trading environments.

From a broader perspective, the study underscores the important role of textual information as a complementary input in electricity markets that are already highly information-rich. Rather than acting as a dominant driver of price levels, news appears to function as a contextual filter that enhances regime identification and directional decision-making. The use of gradient-boosted models at both forecasting and signal-generation stages highlights that the economic value of NLP-based approaches lies not in

maximising predictive complexity, but in disciplined integration with market structure and execution logic.

In conclusion, this research project demonstrates that energy-related news can be transformed into statistically robust and economically meaningful trading signals for the German power market when processed through a carefully designed pipeline combining zero-shot NLP techniques with gradient-boosted machine-learning models. While further methodological refinement and real-time validation are required before practical deployment, the findings provide clear empirical support for the relevance of textual data in short-term electricity trading and contribute to a more integrated understanding of how modern NLP methods, XGBoost-style forecasting, and LightGBM-based signal construction can be combined in applied energy finance research.

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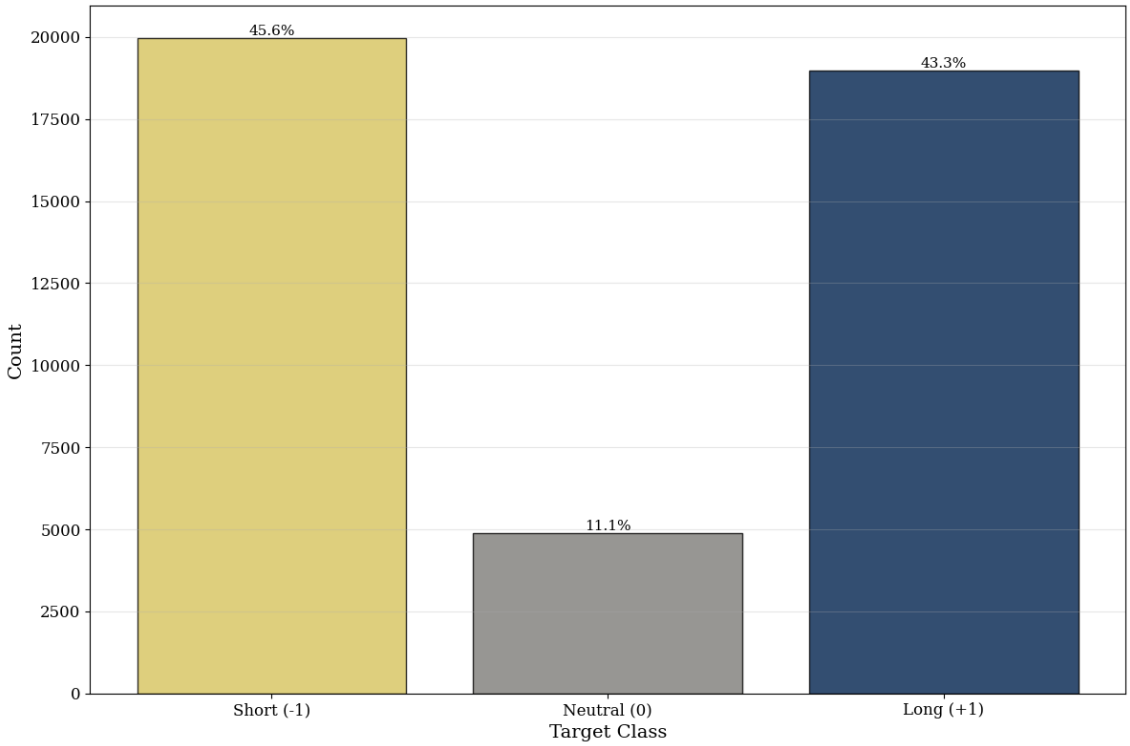
# 8 Appendix

## Appendix 1: Keywords used in News Data Preprocessing

| Category | Keywords used |
|---|---|
| Sports (NFL) | *49ers, bears, bengals, broncos, browns, chargers, chiefs, colts, cowboys, dolphins, eagles, giants, jaguars, jets, lions, packers, panthers, patriots, raiders, rams, ravens, saints, seahawks, steelers, texans, titans, vikings* |
| Sports (NBA) | *nba, basketball, celtics, bulls, lakers, lebron, nets, rockets, sixers, warriors* |
| Sports (Soccer) | *arsenal, celtic, chelsea, cup, league, liverpool, manchester, premier, premiership, scotland, wales* |
| Sports (F1) | *f1, ferrari, formel, hamilton, prix, rennen, verstappen* |
| Sports (MMA) | *champ, makachev, mma, pereira, ufc* |
| Sports (Other) | *champions, championships, coach, college, draft, game, marathon, medal, nhl, olympic, olympics, off-season, roster, rugby, season, sport, swimming* |
| Entertainment | *album, beyonce, concert, disney, drama, espn, fashion, festival, film, hackman, hollywood, kendrick, lamar, lotus, movie, music, netflix, oscar, oscars, pop, songs, star, stream, swift, taylor, tv* |
| Shopping/Retail | *amazon, appliances, best deal, best deals, costco, deals, dyson, grocery, holiday, lululemon, retailer, sale, schnäppchen, shoppers, shopping, walmart* |
| Animals/Wildlife | *abschuss, adorable, animals, bear, cat, dog, dogs, eagle, hund, klauenseuche, puppy, retriever, shark, whale, wolf, wölfe, zoo* |
| Weather/Natural Events | *angeles, arizona, colorado, drought, firefighters, heat, heatwave, hurricane, lake, midwest, snow, storm, storms, summer, tornadoes, weather, wildfire, wildfires* |
| Crime/Violence | *belästigung, drogen, inmate, killing, kokain, messer, murder, police, prison, sexuelle, sexueller, shooting, shot, täter, vergewaltigung* |
| Drugs/Substances | *cannabis, cannabisgesetz, cigarettes, drug, drugs, fentanyl, kiffen, marihuana, marijuana, nicotine, ocerdose, opioid, pills, prescription, vapes, vaping, weed* |
| Space/Science | *asteroid, astronomer, astronomers, eclipse, mars, nasa, space, spacecraft, telescope, webb* |
| Politics/Immigration | *campaign, classified, debate, dei, deportation, deportations, gender, harris, immigrants, immigration, impeachment, kamala, lawsuit, maduro, migrants, senate, students, trans, transgender* |
| Health/Medical | *acne, alzheimer, diabetes, healthy, medicare, medicaid, obamacare, ozempic* |
| Food/Beverage | *alcohol, beef, beer, bourbon, caffeine, chicken, coca, coffee, cola, diet, lobster, meat, soda, starbucks* |
| Real Estate | *estate, homebuyer, homes, housing, mortgage, mortgages, renters* |
| Education | *bildung, education, eltern, harvard, lehrer, schulen, university* |
| Gaming | *brettspiel, fortnite, gamescom, nintendo, playmobil* |
| Gambling | *eurojackpot, jackpot, lotto, lotto24, lottery, powerball* |
| LGBTQ | *lgbt, lgbtq, pride, queere* |
| Horoscope | *horoskop, sternzeichen, wochenhoroskop* |
| Tourism/ Christmas | *glühwein, hotels, tourismus, urlaub, weihnachten, weihnachtsmarkt* |
| Geographic | *bangladesh, congo, ethiopia, gabon, kongo, mali, montana, niger, sudan* |
| Royalty | *charles, king, prince* |
| Labor | *doctors, nurses, strike, uaw, union, workers* |
| Culture | *art, literaturpreis, museum, painting* |
| Media | *podcast, radio, sweeny, sweeney* |
| Other | *7634s, baby, coaching, dad, dance, einsamkeit, family, mom, mutter* |

## Appendix 2: Target Distribution




## Appendix 3: News Distribution per Zero-Shot Topic Classification

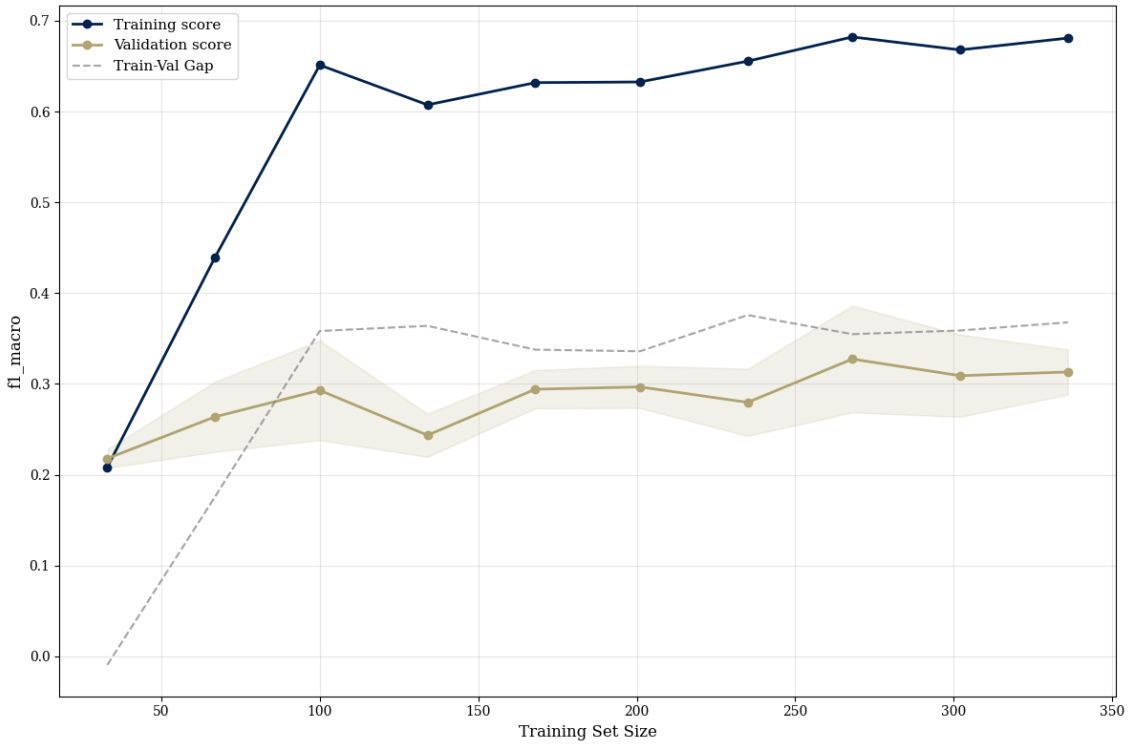
| Zero-Shot Classification | Number of Articles |
|---|---|
| Sport, Unterhaltung, Kultur oder Lokalnachrichten ohne Wirtschaftsbezug | 32’595 |
| Innenpolitik, Regierungsbildung oder Wahlen | 31’234 |
| Krieg, Sanktionen, Terrorgefahr oder geopolitische Spannungen | 19’395 |
| Energiepolitik, Klimagesetze oder EU-Regulierung | 18’314 |
| Technologieentwicklung, Halbleiter, Elektromobilität oder Batterieproduktion | 12’510 |
| Friedensgespräche, Diplomatie oder Aufhebung von Sanktionen | 4’323 |
| Steigende Gas-, Kohle- oder $CO_2$-Preise | 4’314 |
| Zinsentscheidungen, Inflation oder Währungsschwankungen | 3’690 |
| Industriepolitik, Produktionsstandorte oder Unternehmensstrategien | 2’612 |
| Kaltes, windarmes oder bewölktes Wetter | 2’335 |
| Mildes Wetter, starke Winde oder viel Sonneneinstrahlung | 2’001 |
| Fallende Gas-, Kohle- oder $CO_2$-Preise | 1’896 |
| Börsennachrichten, Unternehmensgewinne oder Investitionen | 1’851 |
| Hohe erneuerbare Einspeisung, neue Kapazitäten oder stabile Netze | 1’343 |
| Positive Wirtschaftsentwicklung, steigende Industrieproduktion oder Unternehmenswachstum | 1’020 |
| Rezession, Unternehmenskrise oder sinkende Industrieproduktion | 860 |
| Steigender Stromverbrauch durch Wirtschaft, Industrie oder Extremwetter | 478 |
| Kraftwerksausfälle, Netzengpässe oder geringe erneuerbare Einspeisung | 288 |
| Zölle, Handelskonflikte oder Exportbeschränkungen | 262 |
| Handelsabkommen, Marktöffnung oder Lieferkettenentwicklung | 232 |
| Sinkender Stromverbrauch durch Konjunkturschwäche oder mildes Wetter | 163 |

## Appendix 4: XGBoost Classification Report

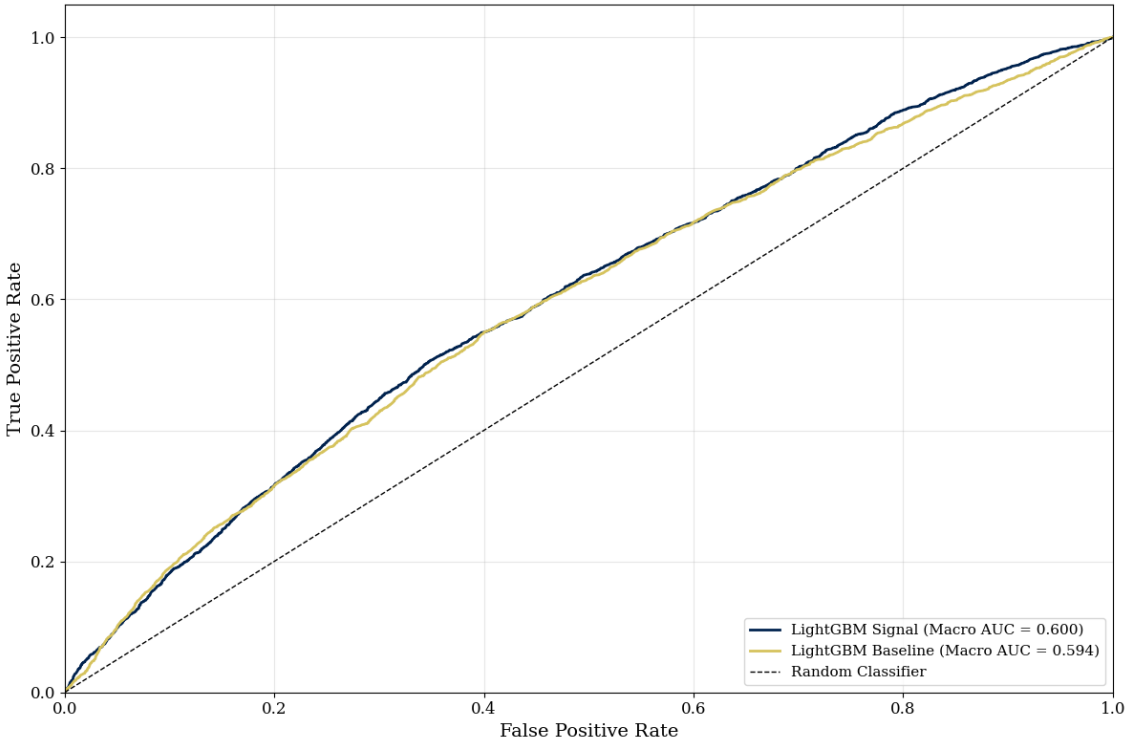
| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Short (-1) | 0.52 | 0.63 | 0.57 | 1866 |
| Neutral (0) | 0.16 | 0.11 | 0.13 | 573 |
| Long (1) | 0.56 | 0.50 | 0.53 | 1942 |

| Accuracy | Macro-F1 | Macro-Averaged AUC |
|---|---|---|
| 0.5049 | 0.4103 | 0.610 |

## Appendix 5: Learning Curve diagnostics



## Appendix 6: Receiver Operating Characteristic Curves for the LightGBM signal and baseline models (test set)




## Appendix 7: Statistical Model Comparison with McNemar & Confidence Intervals

**Panel A: Overall Predictive Accuracy**

| Model | Accuracy |
|---|---|
| Signal Model | 0.4513 |
| Baseline Model | 0.3725 |

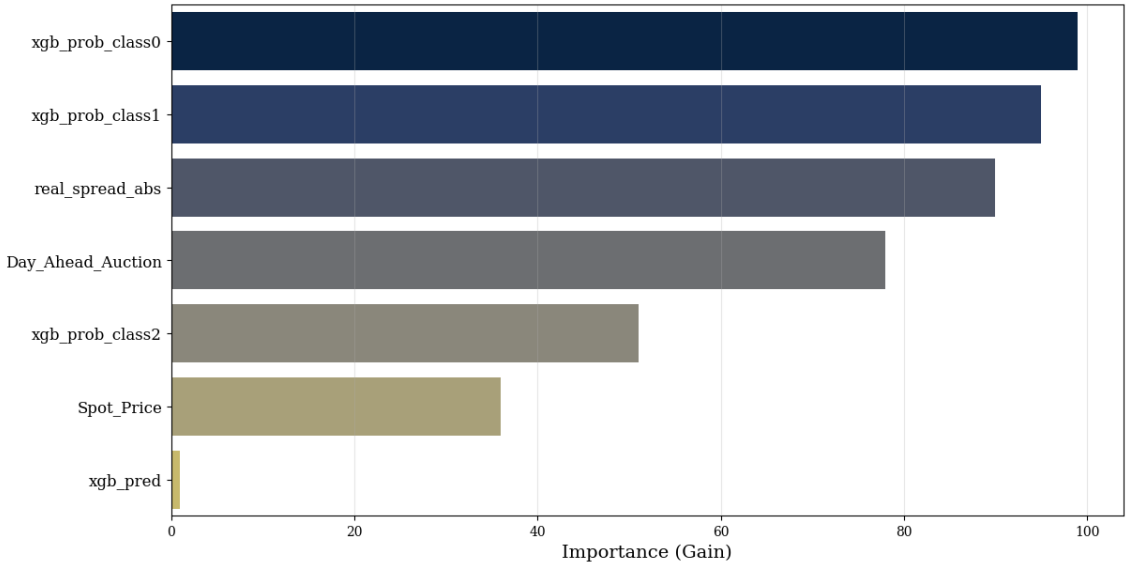
**Panel B: McNemar's Test for Paired Comparison**

| Outcome Category | Count |
|---|---|
| Both correct | 1285 |
| Signal only | 692 |
| Baseline only | 347 |
| Both wrong | 2057 |
| | |
| **Test Statistic** | **Value** |
| McNemar's $\chi^2$ | 113.8941 |
| p-value | 0.0000 |

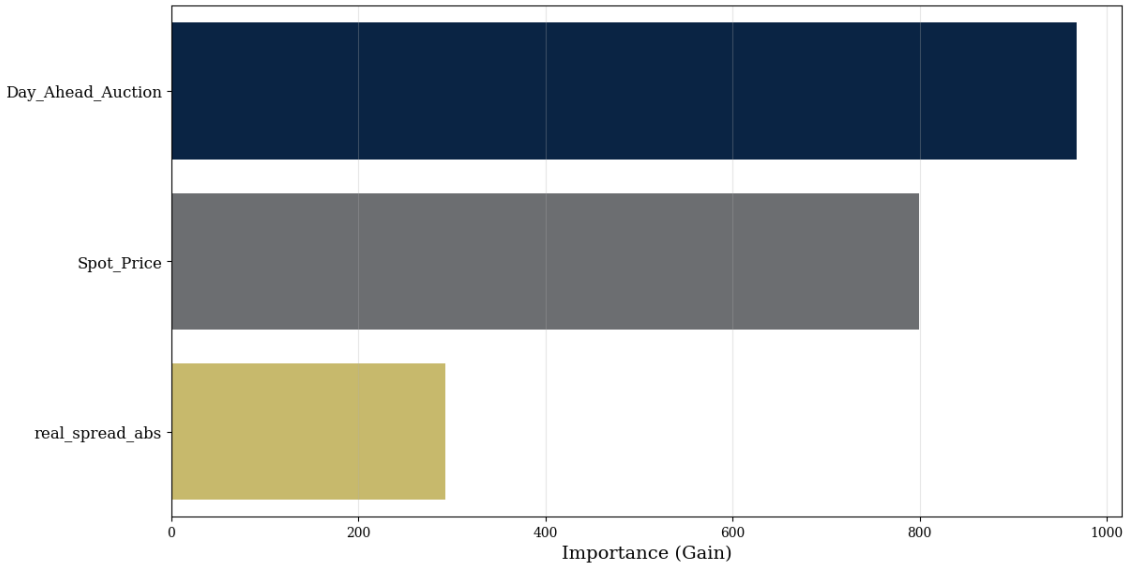
**Panel C: Bootstrap 95% Confidence Intervals for Accuracy**

| | |
|---|---|
| Signal model | 0.4512 [0.4360, 0.4663] |
| Baseline model | 0.3726 [0.3579, 0.3862] |
| Difference | +0.0786 |

## Appendix 8: Features of LightGBM Signal Model




## Appendix 9: Features of LightGBM Baseline Model



**Appendix 10: List of AI tools used**

Anthropic (2025). Claude Code (Opus 4.5, Sonnet 4.5). https://claude.com/

- Helping identify bugs in the codebase
- Expanding markdown cell descriptions in the Jupyter notebook

Anysphere (2025). Cursor (Cursor Tab, Composer V1). https://cursor.com/home

- Auto-completing lines during coding
- Proofing coding syntax
- Execution of code refactors

DeepL (2025). DeepL Translator (Version Free). https://www.deepl.com/write

- Translating and improving text passages

Google (2025). Gemini (Gemini Pro 3). https://gemini.google.com/

- Summary of documentation pages
- Literature research support
- Codebase scanning and debugging

Grammarly (2025). Grammarly AI (Pro Version). https://www.grammarly.com

- Correcting writing and punctuation mistakes
- Rewriting text passages to fit the academic writing style better

OpenAI. (2025). ChatGPT (Version 5.1 / 5.2). https://chat.openai.com/chat

- Thoughts impetus
- Structuring aid for dividing sections in a chapter
- Translating and improving text passages
- Literature research support

GitHub. (2025). GitHub Copilot (Pro Version). https://github.com/features/copilot

- Code inspiration and prototyping
- Debugging and error handling in the code